

Do the Fama and French Factors Proxy for State Variables that Predict Macroeconomic Growth in the Eurozone?*

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Abstract

Liew and Vassalou show that the returns to HML and SMB contain incremental information on future Gross Domestic Product (GDP) growth rates in selected countries. We extend these findings in focusing not only on 16 European countries but also on 11 pan-Eurozone industries and the Eurozone as a whole. In focusing on the latter two, we give a new twist to the Liew and Vassalou methodology since we use it as a means to test whether equity markets across the Eurozone are integrated. Using sample data from January 1990 to April 2008, we find empirical support for the existence of a value, size, and momentum effect in various countries, industries, and in the Eurozone as a whole. In addition, we document that not only the market factor may serve as leading indicator for future real economic activity but also the returns to SMB and to a lesser extent HML. We also link the profitability of WML to future GDP growth rates. We yet fail to find the same significance as for SMB and HML. Our results are in line with an ICAPM explanation for the empirical success of the Fama and French factors. Moreover, we document that the returns to pan-Eurozone and industry specific SMB and HML trading strategies contain incremental information on future GDP growth in the Eurozone. This implies two things. First, industry characteristics have become increasingly important relative to country effects for the explanation of equity returns in the Eurozone. Second, equity markets across the Eurozone are to a certain extent integrated.

Keywords: Asset pricing; Fama-French factors; industry factors; European integration; euro area stock market; GDP growth; diversification

JEL Classification Codes: E44, G12, G15

1 Introduction

Do the Fama and French (1993) factors proxy for state variables that predict macroeconomic growth in the Eurozone? Based on the existing literature and past empirical findings an answer is not yet identifiable. The stated question is, however, of at least a threefold interest.

First, analyzing whether the Fama and French (1993) factors may serve as leading indicators for future macroeconomic growth begs first and foremost the question whether there exist a relationship between equity returns and future real economic activity. In case such a relation exists, a decomposition of equity returns into a size effect and a book-to-market effect as proposed by Fama and French (1992, 1993) may strengthen and enhance any information content of stock returns on future economic development. So far only one study by Liew and Vassalou (2000) has provided empirical evidence that these two firm characteristics contain incremental information on future real economic activities in various countries. This paper intends to advance these findings in (i) focusing on another set of countries and in (ii) studying whether such a relationship between the Fama and French (1993) factors and future GDP growth rates exists also for an economic and monetary union, such as the Eurozone, as well as for different industries.

Second, although the cross-sectional evidence reported by Fama and French (1992, 1993, 1995, 1996, 1998), amongst others, has had a tremendous impact on the area of asset pricing, a fair amount of academic debate has emerged over the economic rationale of using size and book-to-market for the pricing of assets. To date, there is still no clear consensus among economists and financial scholars on whether these two firm attributes may serve as suitable state variables in context of Merton's (1973) Intertemporal Capital Asset Pricing Model (ICAPM). A few studies have already provided some tentative explanations for the economic rationale of the Fama and French (1993) factors, yet with varying success (see Cooper et al., 2001; Ferson and Harvey, 1999; Fama and French, 1996; Heaton and Lucas, 2000; Hodrick and Zhang, 2001; Lettau and Ludvigson, 2001; Liew and Vassalou, 2000; Perez-Quiros and Timmermann, 2000; Petkova, 2006; Vassalou, 2003). This study intends to add to this discussion in providing further response to Cochrane (2005) and Fama (1998), who remark that the ICAPM should not be considered a 'fishing license' for choosing factors that have high explanatory power but intrinsically lack the ability to forecast future investment opportunities.

Third, testing whether pan-Eurozone constructed size and book-to-market factors may serve as suitable state variables within the context of the ICAPM (Merton, 1973) allows more-

over to make inferences on the degree of stock market integration among Eurozone countries. In particular, past studies have shown that the inception of the European Economic and Monetary Union (EMU) and the advent of the euro have had a positive impact on long-run stock market integration (see Abbot and Chow, 1993; Adjaoute and Danthine, 2002; Baele et al., 2004; Danthine et al., 2000; Hardouvelis et al., 2006; Prati and Schinasi, 1997; Serletis and King, 1997; Worthington et al., 2003). Thence, if the integration of stock markets in the Eurozone is due to the economic convergence of the EMU member states, then, in the long-run, equilibrium relations among equity returns in the Eurozone and their variances ought to be affected by economic variables that reflect economic activities undertaken by the EMU. These economic variables, in turn, are exactly those proxied for by potential state variables in context of the ICAPM (Merton, 1973). Here, we test whether the Fama and French (1993) factors may fulfill the role of these state variables. If they do, then this may serve as empirical support for a long-run integration of equity markets across the Eurozone.

Insights on whether the Fama and French (1993) factors contain incremental information on future macroeconomic growth should be of considerable interest to investors and policy makers alike. On the one hand, under the premise that equity markets may serve as leading indicators of real economic activity (see Aylward and Glen, 2000; Binswanger, 2000a,b, 2004; Barro, 1990; Fama, 1981, 1990; Fischer and Merton, 1984; Geske and Roll, 1983; Schwert, 1990; Wahlroos and Berglund, 1986; Wasserfallen, 1989, 1990), policy makers may follow European equity markets and use them as a prompt feedback loop to ascertain whether their convergence efforts may be achieved. On the other hand and under considerations of diversification, if markets appear to be integrated, investors need to decide on whether to invest outside the EMU to enhance their mean-variance frontier, or, alternatively, to find means to diversify EMU-wide if they are reluctant to invest outside the Eurozone.

In order to test the relation between the Fama and French (1993) factors and future macroeconomic growth in the Eurozone, we follow up on the methodology employed by Liew and Vasalou (2000). The latter use multiple least square regressions with future GDP growth rates as dependent variables and *SMB*, *HML*, as well as *WML* (see Carhart, 1997) as explanatory variables. We run these regressions for 16 European countries as well as for the Eurozone as a whole. Besides, under market integration considerations and due to recent empirical findings that suggest the increasing importance of industry effects vis-à-vis country characteristics in pricing equity in the Eurozone, we conduct similar tests along 11 pan-Eurozone industries.¹

¹cf. Urias et al. (1998); Baca et al. (2000); Cavaglia et al. (2000); Diermeier and Solnik (2001); Cavaglia and

Considering individual industries may also be of interest given that some industries are more sensitive to business cycle movements than others. Next to running regressions, we compute the returns to the risk factors during different (i.e., good, bad, and mediocre) states of the business cycle. This allows us to determine to what extent our results may be biased towards the state of the economy.

The remainder of the paper is structured as follows. Section 2 presents a brief review of related literature. Section 3 introduces the research methods of this paper. Section 4 depicts the data set and construction of the portfolios that serve as proxies for a size, book-to-market, and momentum effect. Section 5 provides our empirical results. We conclude in Section 6 with a summary of our findings and suggestions for future research.

2 Literature Review

Empirical results suggest that there exists a predictive relationship between stock market returns and real economic activity. For example, Fischer and Merton (1984) as well as Aylward and Glen (2000) document international evidence that aggregate market returns can be used as leading indicators of future economic growth. Fama (1981, 1990), Geske and Roll (1983), Barro (1990), and Schwert (1990) report that US stock returns are positively related to future macroeconomic increase. Mullins and Wadhvani (1989) find a similar relation pattern for Germany and the United Kingdom. These findings are corroborated by Wahlroos and Berglund (1986) and Wasserfallen (1989, 1990) who identify a positive relation between market returns and future real economic activity for a variety of European countries. Binswanger (2000a,b, 2004) also shows that a substantial fraction of fluctuations in macroeconomic growth can be explained by lagged aggregate stock returns in industrialized countries. This relation is depicted in Figure 1.

Another line of research has focused on the decomposition of asset returns. Albeit the unconditional capital asset pricing model (CAPM) (Sharpe, 1964; Lintner, 1965; Treynor, 1965) is up to now perhaps the most widely used model in asset pricing, it lacks power in explaining the cross-sectional variation of asset returns. Besides, the CAPM is based on a strong set of assumptions. These drawbacks have triggered scholars to derive more advanced asset pricing models. For instance, Merton (1973) extends the CAPM by state variables that

Moroz (2002); Brooks and Catao (2000); L'Her et al. (2002); Wang et al. (2003); Ferreira and Gama (2005); Flavin (2004); Isakov and Sonney (2004); Campa and Fernandes (2006); Moerman (2004, 2008); Taing and Worthington (2005).

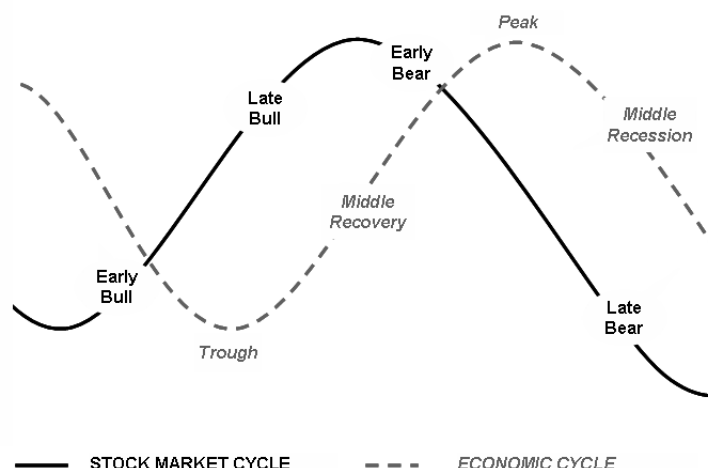


Figure 1: Stock Cycle Leading Economic Cycle

aim at forecasting changes in the investment opportunity set of agents. Ross (1976), on the other hand, proposes an Arbitrage Pricing Theory (APT) model. This relative asset pricing model is based on the law of one price and considers a factor structure for the return generating process. A variety of scholars have, however, criticized the APT for being based on data rather than economic theory (see Black, 1995; Dhrymes et al., 1984; Connor and Korajczyk, 1988).²

Among the enormous number of asset pricing models proposed, the cross-sectional evidence reported by Fama and French (1992, 1993, 1995, 1996, 1998) has had a tremendous impact on the area of asset pricing. Fama and French suggest that their three-factor model (3FM) explains a large percentage of the cross-sectional variation in average returns of portfolios that are sorted by book-to-market and size. The three factors that Fama and French propose are (1) the risk premium of the market portfolio, (2) the return on a portfolio that is long on small stocks and short on big stocks (*SMB*, small minus big), and (3) the return on a portfolio that is long in high-book-to-market stocks and short in low-book-to-market stocks (*HML*, high minus low). Carhart (1997) extends the 3FM incorporating as a fourth factor the return on a portfolio that is long in past winner stocks and short on past loser stocks (*WML*, winner minus loser stocks).

Concatenating (i) the empirical evidences that show that there exists a relation between equity returns and future real economic activity with (ii) the apparent strong explanatory power of the Fama and French (1993) factors in explaining equity returns, begs the question whether size and book-to-market contain explanatory power with respect to future macroeconomic

²A brief history of modern asset pricing literature is presented by Dimson and Mussavian (1999). Campbell et al. (1997) and Cochrane (2005) provide a thorough presentation of the modern theory of asset pricing.

growth as well. It is generally acknowledged that accounting ratios, such as the book-to-market ratio, are supposed to convey growth expectations. In particular, they represent scaled prices with respect to the future. Further, in context of the neoclassical Solow growth model (Solow, 1956), today's economic activities, as e.g., proxied for by today's accounting ratios, condition future macroeconomic growth. Thus, a decomposition of stock returns into multiple factors, each of which convey some sort of growth information, can be expected to reveal incremental information on future economic growth vis-à-vis the one contained in equity market returns themselves. Figure 2 illustrates this simplified relationship between macroeconomic growth in terms of gross domestic product (GDP) growth, equity returns, factor returns, and economic activities.

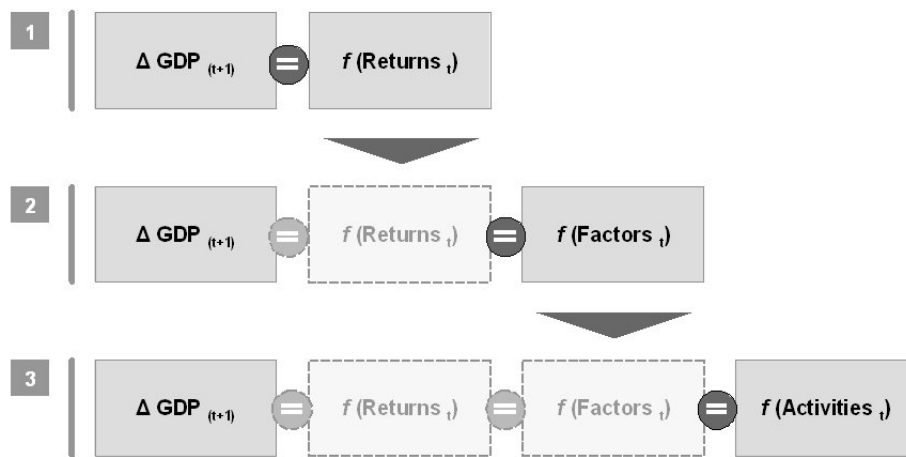


Figure 2: The Relationship Between GDP Growth, Equity Returns, Factors, and Economic Activities

Liew and Vassalou (2000) link value and small-firm returns to macroeconomic events.³ They document that *HML* and *SMB* help to forecast future rates of economic growth in various countries. Albeit Liew and Vassalou (2000) conduct their study for various countries, they do not provide any indication on whether their findings may hold for an economic and monetary union, such as the Eurozone, as well. In other words, the findings of Liew and Vassalou may be biased towards the countries for which the analyses have been conducted. Here we take, however, the position that if the model proposed by Liew and Vassalou (2000) holds in a *pan*-Eurozone context, then *HML* and *SMB* may not only serve as state variables within the context of the ICAPM (see Merton, 1973), but equity markets may also be considered integrated

³Value firms are considered companies that have high-book-to-market ratios; on the other hand, growth firms are companies with low book-to-market ratios

across the Eurozone, given that macroeconomic growth in the Eurozone can be explained by common state variables. This adds a new twist to the methodology proposed by Liew and Vassalou (2000).

Besides, Liew and Vassalou (2000) do not provide any indication to what extent the returns on *HML* and *SMB* may be due to industry specifications captured by the creation of the portfolios that act as their proxies for a size and book-to-market effect.⁴ For instance, it is feasible that stocks that belong to different industries differ in their book-to-market ratios, size, and momentum characteristics. Put differently, by going e.g., long on high book-to-market stocks and short on low book-to-market firms, a *HML* portfolio may contain significantly more stocks of one specific industry than of another.

Although, past studies have remarked that industry portfolios are difficult to price using the conventional CAPM or the Fama and French three factor model (Fama and French, 1997; Moerman, 2004; Van Vliet and Post, 2004), the use of such portfolios, may be considered highly important under the consideration that some industries are more sensitive to business cycle movements than others (Berman and Pfleeger, 1997; Hornstein, 2000; Gourio, 2006). For example, it can be reasonably argued that industries such as utilities, oil, telecommunications, and gas are rather immune to business cycle movements, given that there is a permanent demand for products of these sectors. Yet, other industries, such as retail, and durable consumer goods may be more sensitive to fluctuations in real economic activities. Alternatively, one may state that GDP growth depends on both aggregate and specific industry output, given that some industries have higher correlations with real economic output and development than others.

Besides, taking industry characterizations into account may be especially of interest when considering a pan-European market. First of all, GDP growth in the Eurozone is to a large percentage driven by the two biggest economies in this area, namely Germany and France. Thence, the use of a common area model or country factor models would be strongly biased with respect to the economic activities in these two countries. Secondly, recent findings suggest that industry characteristics have become more important relative to country factors in explaining equity returns throughout Europe (see Urias et al., 1998; Baca et al., 2000; Cavaglia et al., 2000; Diermeier and Solnik, 2001; Cavaglia and Moroz, 2002; Brooks and Catao, 2000; L'Her

⁴In an earlier draft of this paper, Liew and Vassalou (1999) classify their stocks into different industry groups, in order to test whether in all three portfolios (i.e., *HML*, *SMB*, and *WML*) one specific industry seems to be fairly represented in one market as reflected by the number of stocks included in each portfolio. They argue that it is unlikely that the returns of the *HML*, *SMB*, and *WML* portfolios are due to industry characteristics.

et al., 2002; Wang et al., 2003; Ferreira and Gama, 2005; Flavin, 2004; Isakov and Sonney, 2004; Campa and Fernandes, 2006; Moerman, 2008; Taing and Worthington, 2005).

The rationale behind the increasing importance of industry considerations relative to country attributes may lie within the progression of the European Economic and Monetary Union (EMU) and especially the advent of the euro in 1999. The harmonization of monetary, fiscal, and political policies has resulted in a breakup of country borders, while industry barriers have mainly remained stable. Danthine et al. (2000), for instance, remark that the economic convergence has provoked a surge in investments and cross-border trading due to a reduction of implicit and explicit transaction costs and an increased price transparency. This suggests that over the last decade, European investors have become stimulated to hold non-domestic European assets that used to be too costly and risky prior to the arrival of the euro.⁵

To thoroughly determine the factors that persistently drive equity returns in the Eurozone requires the examination of the long-run linkage between economic convergence among EMU member states and the integration of financial markets within the Eurozone. Abbot and Chow (1993), Serletis and King (1997), Prati and Schinasi (1997), and Worthington et al. (2003), amongst others, document that economic integration among EMU member states is a crucial explanation for long-run stock market integration within the European Union. Thus, if the integration of financial markets in the Eurozone is due to the economic convergence of the EMU member states, then, in the long run, any equity prices should be affected by economic variables that reflect economic activities undertaken by the EMU. These economic variables, in turn, are exactly those proxied for by potential state variables in the ICAPM.

The implication of this relationship is that the Fama and French (1993) factors may proxy for state variables within the context of the ICAPM, if there is empirical evidence that size and book-to-market are attributes that contain explanatory power in pricing assets in the Eurozone. This needs to hold since economic theory suggests that stock prices are nothing else than the discounted value of expected future dividends. However, on the reverse, if there was no long-run relationship between the financial markets of the Eurozone and the economic convergence of the EMU, then the Fama and French model may rather be considered an asset pricing model in the context of the Arbitrage Pricing Theory (APT) Ross (1976). However, no previous study has performed necessary tests to validate this statement.

⁵Eijffinger and Lemmen (1995); Guiso et al. (2004); Adjaoute and Danthine (2002); Baele et al. (2004), and Hardouvelis et al. (2006) provide detailed discussions on regulatory changes.

3 Research Methodology

The model proposed by Liew and Vassalou (2000) finds its origin in the Carhart (1997) four factor model, which is simply an extension of the Fama and French (1993) model by a momentum effect. The Carhart (1997) four factor model is defined as follows:

$$R_{it} - R_{ft} = \alpha_{it} + \beta_{it}MRF_t + h_{it}HML_t + s_{it}SMB_t + w_{it}WML_t + \epsilon_{it} \quad (1)$$

where R_i is the return on an asset i , R_f is the return on a risk free asset, MRF denotes the market risk factor, i.e., the market premium, HML is the return to a portfolio that is long in high-book-to-market stocks and short in low-book-to-market stocks (the so-called *value* effect), SMB is the return to a portfolio that is long on small stocks and short on big stocks (the so-called *size* effect), and WML is the return to a portfolio that is long in past winner stocks and short on past loser stocks (the so-called *momentum* effect). α denotes the pricing error, also referred to as Jensen's alpha (Jensen, 1968). β , h , s , and w represent factor loadings. The error term is denoted ϵ_i and is assumed to follow a white noise process.⁶

Now, rather than considering the returns to equity portfolios as dependent variables (see Carhart, 1997; Fama and French, 1992, 1993, 1995, 1996, 1998), we regress in multiple steps macroeconomic growth, as proxied by GDP growth, on the lagged returns of our four factors. This may allow us to infer whether the Fama and French (1993) factors, along with momentum, may serve as state variables within the context of Merton's Intertemporal Capital Asset Pricing Model (ICAPM) (Merton, 1973). Yet, as we are primarily interested in testing whether there are common state variables that predict *pan-Eurozone* macroeconomic growth, we focus our analyses not only on individual countries, but also on industries across the Eurozone. Especially the latter case is of interest under consideration of equity market integration. If we find significant relationships between our risk factors and future macroeconomic growth in the Eurozone, then we cannot only make an inference about the qualification of our risk factors as state variables; even more interesting from our point of view is the fact that significant relationships provide empirical evidence for the integration of equity markets across the Eurozone.

⁶A variable is said to be white noise if it has zero mean, constant variance, and all of its autocovariances are zero.

3.1 The Relation Between Risk Factors & the Macroeconomy

3.1.1 The One Factor Model

In line with Liew and Vassalou (2000), we start our analyses with a simple regression of future GDP growth on the lagged return to each of the individual risk factors. This allows us to test the explanatory power of each individual regressor with respect to future macroeconomic growth. In particular,

$$GDP_{growth(t,t+4)} = \alpha + \beta FactorRet_{t-4,t} + \epsilon_{t,t+4} \quad (2)$$

where GDP_{growth} is the continuously compounded GDP growth rate per country or for the Eurozone one period hence. $FactorRet$ represents either MRF , HML , SMB , or WML . The residual term of the regression is given by ϵ . The four quarter (i.e., one year) time lag between GDP_{growth} and the independent variables is required in order to test for the prediction of real activity based on current (or previous) risk factor returns.

Based on the results of Liew and Vassalou (2000) and Moerman (2004) we expect that the performances of MRF , HML , SMB , and WML are positively related to future economic growth. This holds for both, regressions by country and industry. In general, a positive relation exists, if high returns to HML , SMB , and WML are associated with a GDP growth in the euro area. More specifically, high book-to-market, small capitalization, and past winner stocks are better able to thrive than low book-to-market, big capitalization, and past loser stocks in times of economic upturns.

As GDP growth rates are observed at quarterly frequencies, successive annual growth rates have three overlapping quarters. This causes autocorrelation among the residuals of equation 2. We correct for the presence of autocorrelation and heteroscedasticity of the error terms using the Newey and West (1987) estimator, setting the lags equal to three.

3.1.2 Multifactor Regressions

Next to univariate regressions, we also run multiple regressions to allow for the comparison of the various additional risk factors relative to the excess market return. In particular, it enables us to test for information content in a variable additional to that contained already in the market factor MRF . In a first step, and again analogous to Liew and Vassalou (2000), we use two-factor regressions that simultaneously include the market factor MRF , and the return

on one trading strategy, i.e., *FactorRet*. It follows

$$GDP_{growth(t,t+4)} = \alpha + \beta MRF_{t-4,t} + \gamma FactorRet_{t-4,t} + \epsilon_{t,t+4} \quad (3)$$

where *FactorRet* is either *HML*, *SMB*, or *WML*. Again, even by incorporating the market excess return as a permanent factor, we expect that the other coefficients remain positive, implying positive relationship with economic growth. In addition, the adjusted R^2 s are expected to be higher than those of the univariate analysis (see Liew and Vassalou, 2000). If the slope coefficient γ turns out to be non-zero at a significant level, the additional risk factors contain information about the future state of the macroeconomy that cannot be fully explained by the market factor itself.

Finally, using in a next step multi-factor regressions that includes variables side-by-side may provide further insights about the marginal benefit of adding new risk factors, given the joint slope coefficients of the regressors. We employ, in particular,

$$GDP_{growth(t,t+4)} = \alpha + \beta MRF_{t-4,t} + hHML_{t-4,t} + sSMB_{t-4,t} + \epsilon_{t,t+4} \quad (4)$$

for the Fama and French (1993) three factor model, and

$$\begin{aligned} GDP_{growth(t,t+4)} &= \alpha + \beta MRF_{t-4,t} + hHML_{t-4,t} \\ &+ sSMB_{t-4,t} + wWML_{t-4,t} + \epsilon_{t,t+4} \end{aligned} \quad (5)$$

for the Carhart (1997) four factor model.

If any added variable comes along with a significant non-zero factor loading and increased adjusted R^2 , which accounts for degrees of freedom, then it may be worthy to include this variable in the regression; put differently, this variable may be considered a state variables in context of Merton's ICAPM (see Merton, 1973) as it allows to make inference about the state of the economy not contained in the market factor itself. Besides, for pan-Eurozone industries significant non-zero factor loadings per risk factors may serve as indicators for European stock market integration.

4 Data & Descriptive Statistics

4.1 Sample Period and Data Sources

Our sample includes monthly data ranging from January 1990 to April 2008. These data, such as firms' equity prices, indices, interest rates, and firm ratios are collected through Datastream. In particular, beginning of month equity prices and indices are taken from Datastream's TOTMK indices. We also include the DJ EuroStoxx index in our analyses whenever we refer to pan-European indices. The return on a one-month ecu-market deposit serves as our risk-free return. Equity prices are adjusted for stock splits and dividends. Quarterly GDP growth rates are obtained per country and for the Eurozone from the Organization for Economic Cooperation and Development (OECD) data warehouse. Monthly firm data have been adjusted to meet the frequency of quarterly GDP growth rates.⁷

We choose a monthly frequency for the firm data since it accounts for speed in arbitrage adjustments but mitigates any potential problems that are associated with microstructure issues such as bid-ask spreads. Besides, the use of monthly data allows us to neglect that there might be no simultaneous trading for a given day, as trading days may differ per country, e.g., due to local bank holidays.

Each firm considered is classified by country and industry. We draw our sample for the 12 EMU member states as of January 2006, i.e., Austria, Belgium, Finland, France, Germany, Greece, Ireland, Italy, Luxembourg, the Netherlands, Portugal, and Spain.⁸ These countries comprise our *Eurozone*. In addition, we extend our sample for robustness analyses by three further members of the European Union (EU), i.e., Denmark, Sweden, and the United Kingdom, plus two other European countries, i.e., Norway and Switzerland. The Eurozone countries plus Denmark, Sweden, and the United Kingdom comprise our *European Union* sample. Eventually, these EU countries plus Norway and Switzerland make up our common *European* market. Smaller countries are usually ignored for these kind of studies due to the short number of stocks available. Overall, the availability of data and the number of firms differ considerably per country. Figure 3 illustrates the distribution and the time window for which data is available. In general, a disadvantage of using European data as opposed to US data is the amount of data available per stock. Overall, the number of stocks may vary due to new stock issues,

⁷We use a monthly (rather than quarterly) data set for firm data as we rely on a common data warehouse whose data we also employ in two other studies.

⁸We do not include the EMU member states Cyprus, Malta (both as of January 2008), and Slovenia (as of January 2007) in our analyses, simply due to limitations of data availability and a potential lack of market integration.

mergers, takeovers, and bankruptcies, or simply due to a lack of data availability.

For firms in the Eurozone, prices are given in euros. Prior to January 1999, prices are given in ecu, which is in accordance with Datastream computations. For non-members of the EMU, we compute prices and returns based on the countries' respective exchange rate with either (i) the ecu prior to 1999 or (ii) the euro as of 1999.

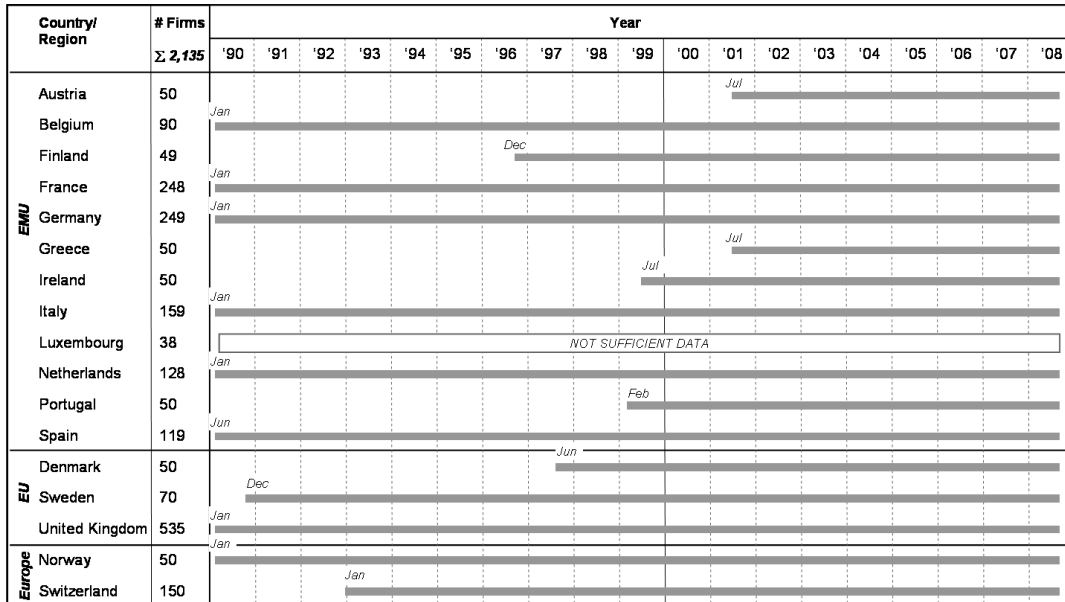


Figure 3: Overview Data Availability per Country/Region

We also classify the firms in our sample along ten different industries as defined by the *Financial Times Actuaries*. These industries include: *basic industries*, *cyclical consumer goods*, *cyclical services*, *financials*, *general industries*, *information technology*, *non-cyclical consumer goods*, *non-cyclical services*, *resources*, and *utilities*.⁹ Besides, for further analytical purposes, we group the industries *cyclical services*, *non-cyclical services*, and *financials* under the common umbrella *services*. The remaining industries are clustered under *industries*. Again, the availability of data and the number of firms differ per industry/service. This is depicted in Figure 4.

Figure 5 provides a joint overview of the distribution per country and industry. As countries like Germany and France, which are the largest countries in the Eurozone, have the highest proportion of stocks in our data sample (neglecting the United Kingdom), one could perhaps argue that some industries, such as *resources* and *basic industries*, are to some extent country specific, since they comprise only a few stocks of smaller countries, such as Greece or Ireland. Yet, given the increasing importance of industry factors vs. country factors in the Eurozone,

⁹Please refer to <http://www.ftse.com> for a detailed description of this classification.

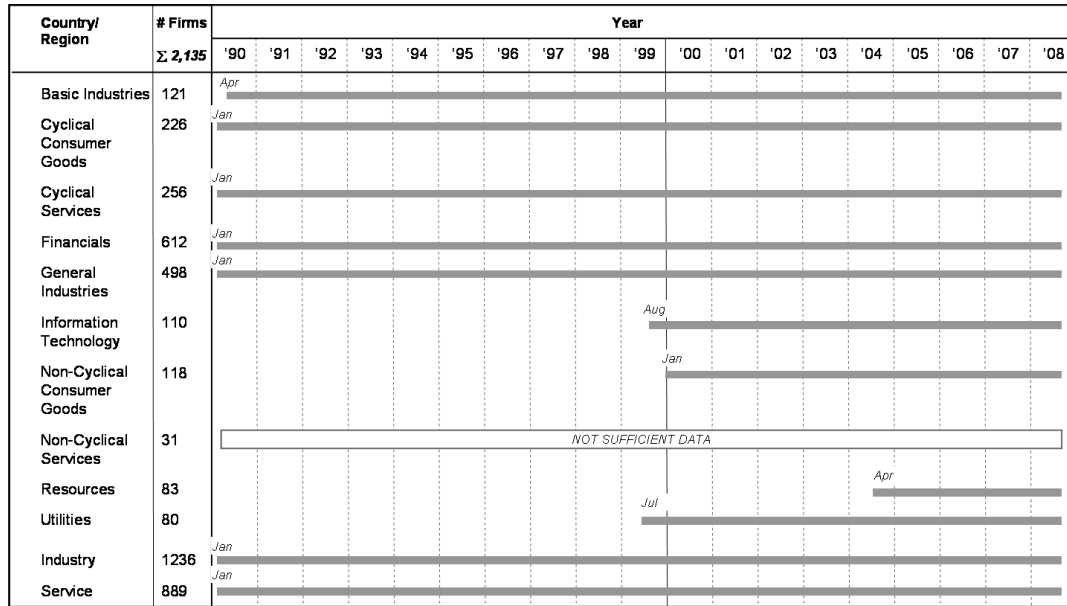


Figure 4: Overview Data Availability per Industry

we consider it appropriate to cluster our firms along the two dimensions, i.e., country and industry. Nevertheless, the interpretation of the empirical results needs to take into consideration whether some industries might be biased towards one specific country. If this is the case, then the industry factor may actually turn out to be a country factor.

All in all, based on the previous discussion on the degree of market integration in the euro area, the selection of the sample period depicts a dilemma. The shorter the time period, the higher the probability that a country (industry) might be fairly underrepresented relative to other countries (industries) as less data become available. As a consequence, the shorter the time period, the lower becomes the number of stocks per country (industry); in turn, the lower the explanatory power and reliability of the data set. On the other hand, as the first step of the European Economic and Monetary Union was just officially launched in 1990, implementing data way prior to this date may seem inappropriate under market integration considerations. In other words, there exists a trade-off between the availability of data and the compliance with the null hypothesis of integrated markets.

4.2 Portfolio Construction

For the construction of the risk factors *HML*, *SMB*, and *WML*, we follow Liew and Vassalou (2000) rather than Fama and French (1993) or Carhart (1997). The Liew and Vassalou (2000) approach differs to some extent from Fama and French's classification. The former employ

Country / Region	Industry / Services											
	Cyclical				Non-Cyclical				Total			
	Basic Industries	Consumer Goods	Cyclical Services	Financials	General Industries	Information Technology	Consumer Goods	Non-Cyclical Services	Resources	Utilities	Services	Industries
Europe	Austria	4	3	2	19	16	0	1	2	2	22	28
	Belgium	6	11	8	29	16	5	2	1	2	39	51
	Finland	8	4	8	5	17	2	1	1	1	14	35
	France	8	35	37	54	57	16	4	8	11	95	153
	Germany	18	31	21	57	67	17	2	11	10	80	189
	Greece	4	6	4	17	9	0	2	2	3	23	27
	Ireland	2	10	7	8	10	4	0	5	0	15	35
	Italy	6	22	14	50	35	7	3	4	14	67	92
	Luxembourg	2	7	5	19	1	0	0	1	3	24	14
	Netherlands	5	12	17	32	39	14	2	2	0	51	77
	Portugal	5	3	12	6	15	3	2	1	2	20	30
	Spain	9	15	13	31	24	4	1	4	10	45	74
	Denmark	0	7	0	19	12	1	0	1	1	19	31
	Sweden	5	7	8	21	20	3	2	1	0	31	39
	United Kingdom	26	35	87	193	113	26	7	21	10	287	248
	Norway	3	5	1	4	15	1	1	17	3	6	44
	Switzerland	10	13	12	48	32	7	1	1	8	61	89
Total	121	226	256	612	498	110	118	31	83	80	899	1236
												2135

Figure 5: Classification of Firms by Country and Industry

a smaller sample size and hence use a three sequential sort rather than two independent classifications that created the *HML* and *SMB* trading strategies of the latter. Albeit Liew and Vassalou (2000) mention that their results may therefore be specific to the classification and sorting they use, they run a robustness test which shows that their results are stable. Given our small sample size at hand, especially per industry, we decide to follow Liew and Vassalou's approach rather than Fama and French's independent sorting.¹⁰

The construction considers all the three anomalies (i.e., *SMB*, *HML*, and *WML*) simultaneously. In order to calculate the momentum of a stock, we only use stocks for which we are able to derive the market capitalization of at least twelve months in a row. As previously mentioned, we incorporate only a twelve month momentum strategy. The momentum strategy is implemented by deriving the mean of past year's returns, excluding however the most recent month.¹¹ Winners are defined as the top third of the stocks per country (or the top third per industry) with the highest last year's average return. Correspondingly, losers comprise the bottom third per country (or per industry, respectively). The medium stocks are the remaining (middle) third of the sample.

To build the risk factors, we sort all stocks by their book-to-market ratio for each month in year $t-1$, and classify them into three different groups, i.e., high, medium, and low book-to-market ratios.¹² We then take each of these portfolios and re-sort all stocks according to their market capitalization (i.e., small, medium, and big market capitalization). Thereby, three portfolios within each book-to-market group are created. This leads to nine portfolios. In a next step, those nine portfolios are again subdivided based on each stock's momentum (i.e., losers, middle group, and winners). Eventually, we obtain 27 portfolios, which are numbered $P1$ through $P27$.¹³ Table 1 provides an overview on the portfolio construction.

¹⁰With respect to our portfolio construction procedure, which differs from that of Fama and French (1993) and which uses a three sequential sort rather than two independent sorts, one may argue that our results are specific to the sorting order used. However, due to the small number of stocks per industry, we cannot use independent sorting. Thus, in order to test the robustness of our results, one could reverse the sorting order and repeat the analyses.

¹¹Liew and Vassalou (2000) suggest to exclude the most recent month in order to eliminate problems that are associated with microstructure issues such as the bid-ask spread.

¹²For the book-to-market ratio we use the inverse of the Datastream price-to-book value datatype (BP). Book value refers thereby to the latest book value shown on the balance sheet.

¹³Since we create 27 portfolios, the number of securities has to be at least 27. If one country/industry has more than 27 stocks, then we first divide the total number of stocks in this country/industry by 3. The greatest feasible divisor is then included in the extreme portfolios, i.e., high/low (for book-to-market), small/big (for size), and winner/loser (for momentum). The remaining stocks are sorted in the respective middle portfolio. For instance, in our sample, the total number of stocks for Spain is 119. After having ranked these stocks by their book-to-market ratio, we divide 119 by 3 and obtain 39.6666. We thus put the 39 stocks with the highest book-to-market ratio into the first portfolio that will thus include all *value* stocks. The lowest ranked 39 assets are put into the portfolio with the assets comprising the lowest book-to-market ratio. The remaining 41 [= 119 - 39 - 39] stocks are then put in the middle portfolio. We follow the same logic for the remaining

Table 1: Portfolio Construction

This table shows the portfolio construction procedure which is in accordance with Liew and Vassalou (2000).			
Book-to-market	Market capitalization	Past year's returns (excl. last month)	Portfolio
High	Small	Losers	P1
		Medium	P2
		Winners	P3
	Medium	Losers	P4
		Medium	P5
		Winners	P6
	Big	Losers	P7
		Medium	P8
		Winners	P9
Medium	Small	Losers	P10
		Medium	P11
		Winners	P12
	Medium	Losers	P13
		Medium	P14
		Winners	P15
	Big	Losers	P16
		Medium	P17
		Winners	P18
Low	Small	Losers	P19
		Medium	P20
		Winners	P21
	Medium	Losers	P22
		Medium	P23
		Winners	P24
	Big	Losers	P25
		Medium	P26
		Winners	P27

These 27 portfolios represent the ingredients for the three risk factors, i.e., *HML*, *SMB*, and *WML*. This is in line with the existing literature. In particular, the factors are computed by adding and subtracting the returns on the *individual* portfolios as follows:

$$HML = 1/9 * \left[\begin{array}{l} (P1 - P19) + (P2 - P20) + (P3 - P21) + (P4 - P22) + (P5 - P23) \\ + (P6 - P24) + (P7 - P25) + (P8 - P26) + (P9 - P27) \end{array} \right]$$

$$SMB = 1/9 * \left[\begin{array}{l} (P1 - P7) + (P2 - P8) + (P3 - P9) + (P10 - P16) + (P11 - P17) \\ + (P12 - P18) + (P19 - P25) + (P20 - P26) + (P21 - P27) \end{array} \right]$$

$$WML = 1/9 * \left[\begin{array}{l} (P3 - P1) + (P6 - P4) + (P9 - P7) + (P12 - P10) + (P15 - P13) \\ + (P18 - P16) + (P21 - P19) + (P24 - P22) + (P27 - P25) \end{array} \right]$$

rebalancing steps.

In summary, *HML* describes the return on a portfolio that is long on high book-to-market firms and short on low book-to-market firms. By simultaneously controlling for *SMB* and *WML*, *HML* becomes size and momentum neutral. Accordingly, *SMB* and *WML* are corrected for a book-to-market and momentum or size effect. This approach allows us therefore also to eliminate any problems of potential multicollinearity among the risk factors.¹⁴

The individual risk factor returns are derived for quarterly, semi-annual, and annual rebalancing frequencies for equally weighted portfolios per country, per industry across our Eurozone countries, as well as for the Eurozone as a whole.

4.3 Descriptive Characteristics of the *HML*, *SMB*, and *WML* strategies

Before concatenating the risk factors with future GDP growth, we study briefly their performance and characteristics per country, industry, and for the Eurozone. Tables 2 and 3 report the summary statistics for annually rebalanced portfolios that serve as ingredients for the risk factors *HML*, *SMB*, and *WML* per country and industry, respectively. Table 2 also depicts summary statistics for GDP growth rates per country and for the Eurozone.

The statistics shown are derived using all data available per country and industry. As such, as indicated in section 4.1, the time periods, and hence the number of observations, might differ per country/industry. Although we intend to focus our discussion primarily on annually rebalanced portfolios to be in line with the existing literature, two quick notes on higher turnover frequencies appear to be worth noting.¹⁵

¹⁴Albeit the presence of multicollinearity does not affect the consistency of ordinary least squares estimates of the regression coefficients, the estimates become extremely imprecise and unreliable. Besides, distinguishing the individual impacts of the independent variables on the dependent variables becomes practically infeasible. The statistical consequence is presented in inflated standard errors for the factor loadings of the regression. We also derived variance inflation factors (VIF) (see Wooldridge, 2000) per risk factor and country as well as industry. When following the common rule of thumb, i.e., a $VIF(\hat{\beta}_i) > 10$ is said to imply high multicollinearity (see Kutner et al., 2003), our results suggest that there is no severity in multicollinearity among the risk factors. The results are depicted in Tables I and II in the Appendix.

¹⁵The findings for quarterly and semi-annually frequencies are not presented here due to space considerations. The results can, however, be requested from the authors.

Table 2: Summary Statistics per Country and Eurozone

This table reports the annualized summary statistics for GDP growth rates as well as for all risk factors considered per country and the Eurozone. The countries are clustered along three dimensions. The first group comprises those countries that belong to the Eurozone. The second cluster represents countries of the European Union that do not belong to the Eurozone. The last cluster contains European countries that neither belong to the Eurozone nor the European Union. The results are based on annually rebalanced HML, SMB, and WML portfolios. MRF denotes the market risk factor. HML is the return on a portfolio that is long on high book-to-market stocks and short on low book-to-market securities, holding size and momentum characteristics of the portfolio constant. SMB is the return on a portfolio that is long on small capitalization stocks and short on big capitalization securities, holding book-to-market and momentum characteristics of the portfolio constant. WML is the return on a portfolio that is long on the best performing stocks of the past year ('winners') and short on the worst performing securities of the previous year ('losers') holding book-to-market and size characteristics of the portfolio constant. GDP refers to GDP growth rates in the respective countries or the Eurozone. *, **, *** used for the Jarque-Bera (JB) test and for the Augmented Dickey Fuller (ADF) test denote, respectively, significance at the 10%, 5%, and 1% significance level.

	Mean (%)	Median (%)	Std. (%)	Skweness	Kurtosis	Jarque-Bera	ADF
Austria							
MRF	18.89	14.35	19.83	0.295	1.642	2.715	-1.254
HML	8.26	4.09	20.41	1.291	4.780	8.600**	-1.555
SMB	11.73	9.96	20.77	0.481	2.047	2.218	-2.566
WML	4.86	5.21	10.93	-0.478	3.729	1.098	-2.413
GDP	3.83	4.05	1.39	-0.034	1.419	3.093	-1.744
Belgium							
MRF	9.94	12.79	21.37	-0.396	2.169	2.820	-1.997
HML	5.12	7.26	12.82	-0.636	2.843	3.120	-2.920*
SMB	10.34	9.42	12.90	0.248	2.248	1.855	-2.999**
WML	7.73	6.25	14.44	0.318	2.053	2.835	-2.832*
GDP	3.95	4.26	1.37	-0.292	2.008	2.893	-3.489**
Finland							
MRF	23.01	21.50	48.13	0.724	3.691	4.090	-2.491
HML	23.58	11.33	53.07	3.499	16.341	375.146***	-3.674***
SMB	28.65	13.73	55.46	3.100	14.393	277.488***	-4.021***
WML	1.18	0.79	10.72	-0.004	3.476	0.187	-2.678*
GDP	5.47	5.68	2.46	0.063	2.157	1.648	-2.359
France							
MRF	8.41	10.24	25.33	0.161	2.660	1.127	-4.737***
HML	10.97	5.36	24.81	2.398	11.153	381.172***	-3.847***
SMB	9.56	9.75	19.95	-0.266	3.083	1.226	-4.015***
WML	4.51	2.65	13.66	1.232	5.270	47.337***	-5.486***
GDP	5.47	4.44	3.29	1.381	4.818	46.373***	-3.653***
Germany							
MRF	6.10	11.83	21.98	-0.317	2.551	1.768	-3.178**
HML	10.44	7.05	17.75	1.814	7.635	85.077***	-4.694***
SMB	13.53	4.56	24.75	1.689	5.958	49.524***	-2.438
WML	6.58	5.45	8.67	0.511	3.228	2.697	-3.816***
GDP	2.91	2.48	1.66	1.158	4.223	16.821***	-3.948***
Greece							
MRF	5.10	13.82	25.57	-0.254	1.699	2.381	-1.649
HML	12.41	9.25	23.92	0.257	2.218	1.196	-2.275
SMB	15.61	1.43	33.52	0.846	2.591	3.026	-1.116
WML	0.47	2.59	21.62	0.411	3.995	1.106	-3.481**
GDP	7.71	7.77	0.78	-0.182	2.198	1.111	-1.899
Ireland							
MRF	4.52	7.93	16.81	-0.454	2.390	1.814	-1.407
HML	25.42	16.29	32.25	1.927	6.973	36.024***	-2.670*
SMB	10.15	1.46	31.93	1.086	4.170	7.077**	-2.929*
WML	-3.95	3.02	28.93	-1.325	6.498	22.107***	-3.608**
GDP	9.88	9.07	4.39	0.808	3.602	3.479	-2.427
Italy							
MRF	3.63	4.62	24.47	0.429	3.317	2.472	-2.954**
HML	5.14	3.77	13.37	-0.121	2.916	0.265	-4.591***
SMB	6.41	6.34	15.94	-0.062	3.488	0.565	-3.399**
WML	3.84	3.73	10.92	-0.318	3.447	1.682	-5.468***
GDP	5.57	4.66	2.73	0.723	2.241	8.692**	-2.669*
Netherlands							
MRF	5.73	8.05	20.69	0.134	3.718	1.537	-3.030**
HML	4.58	1.00	16.30	0.875	3.978	12.401***	-3.806***
SMB	6.74	4.59	17.45	0.558	3.087	3.994	-3.081**
WML	3.83	2.50	13.06	-0.126	3.104	0.206	-4.147***
GDP	5.07	5.14	1.62	0.145	2.605	0.974	-2.397
Portugal							

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Table 2 – continued from previous page

	Mean (%)	Median (%)	Std. (%)	Skweness	Kurtosis	Jarque-Bera	ADF
MRF	1.13	3.58	21.68	-0.139	1.564	3.491	-1.856
HML	18.07	7.58	31.11	1.995	7.989	51.329***	-3.627**
SMB	4.41	-1.80	37.10	2.320	9.621	82.661***	-3.087**
WML	-0.85	-2.28	17.19	0.131	4.210	1.470	-2.876*
GDP	4.75	4.56	1.59	0.144	2.406	0.886	-1.677
Spain							
MRF	13.57	14.68	23.13	0.058	2.570	0.601	-1.734
HML	10.91	13.40	17.72	-0.238	3.904	1.530	-2.941*
SMB	16.12	4.51	26.97	1.273	4.206	14.298***	-2.501
WML	-0.69	0.72	18.77	-1.022	7.142	37.455***	-3.896***
GDP	7.37	7.39	0.89	-0.539	3.226	2.196	-2.492
Denmark							
MRF	12.79	14.85	23.74	-0.192	2.222	1.604	-2.941*
HML	14.63	16.06	18.17	0.158	3.175	0.164	-4.354***
SMB	21.21	9.95	27.32	0.910	3.134	5.392*	-2.094
WML	-2.24	-1.04	16.84	-0.178	3.457	0.359	-3.176***
GDP	4.23	4.08	1.85	-0.217	2.242	1.611	-2.678*
Sweden							
MRF	16.91	19.42	33.22	0.441	3.907	3.101	-3.207**
HML	14.08	7.22	37.98	3.913	19.230	699.514***	-5.403***
SMB	12.57	12.37	22.88	-0.214	3.300	0.481	-2.544
WML	-4.01	-3.03	21.40	-2.296	9.740	142.028***	-5.512***
GDP	4.91	4.63	1.69	-0.105	3.229	0.128	-2.643**
United Kingdom							
MRF	5.86	6.55	14.84	-0.402	3.278	3.006	-3.871***
HML	5.35	5.26	9.73	0.345	4.269	8.370**	-4.824***
SMB	10.13	7.64	14.24	1.769	8.762	194.023***	-4.964***
WML	2.26	2.56	9.54	-0.892	4.667	24.931***	-5.938***
GDP	6.84	6.18	2.16	0.721	2.568	10.049***	-2.189
Norway							
MRF	12.17	9.97	28.66	0.335	2.336	3.176	-3.738***
HML	5.29	3.21	18.06	1.021	5.238	28.196***	-5.191***
SMB	2.80	3.70	18.40	0.071	5.079	12.771***	-3.806***
WML	3.73	2.27	18.05	0.065	3.370	0.326	-4.756***
GDP	6.83	6.46	5.13	0.706	4.026	9.254***	-3.110**
Switzerland							
MRF	10.04	10.82	20.94	-0.103	2.697	0.484	-3.004**
HML	12.22	12.61	31.25	-0.037	3.309	0.113	-2.573
SMB	15.05	8.68	27.53	1.104	4.531	16.348***	-3.432**
WML	-2.75	2.19	21.66	-2.064	9.056	123.345***	-3.988***
GDP	2.47	2.13	1.66	0.351	1.991	4.055	-2.875*
Eurozone							
MRF	5.73	6.82	22.30	-0.239	2.469	1.820	-3.377**
HML	6.91	6.96	8.38	0.224	2.617	1.260	-4.583***
SMB	11.34	10.58	12.56	0.555	3.825	5.414*	-2.960**
WML	4.04	4.04	9.08	-1.255	6.092	46.405***	-4.793***
GDP	4.87	4.61	1.74	0.290	2.762	1.330	-3.152**

Table 3: Summary Statistics per Industry (Eurozone)

This table reports the annualized summary statistics for all risk factors considered per industry. The results are based on annually rebalanced HML, SMB, and WML portfolios. MRF denotes the market risk factor. HML is the return on a portfolio that is long on high book-to-market stocks and short on low book-to-market securities, holding size and momentum characteristics of the portfolio constant. SMB is the return on a portfolio that is long on small capitalization stocks and short on big capitalization securities, holding book-to-market and momentum characteristics of the portfolio constant. WML is the return on a portfolio that is long on the best performing stocks of the past year ('winners') and short on the worst performing securities of the previous year ('losers') holding book-to-market and size characteristics of the portfolio constant. *, **, *** used for the Jarque-Bera (JB) test and for the Augmented Dickey Fuller (ADF) test denote, respectively, significance at the 10%, 5%, and 1% significance level.

	Mean (%)	Median (%)	Std. (%)	Skweness	Kurtosis	Jarque-Bera	ADF
Basic Industries							
MRF	5.97	6.27	21.77	-0.221	2.680	1.028	-2.900*
HML	12.13	5.64	22.33	1.108	3.888	15.945***	-3.280**
SMB	4.33	-1.31	25.71	1.048	4.427	17.746***	-3.255**

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Table 3 – continued from previous page

	Mean (%)	Median (%)	Std. (%)	Skweness	Kurtosis	Jarque-Bera	ADF
WML	1.62	2.12	17.31	-0.879	7.610	66.970***	-5.928***
Cyclical Consumer Goods							
MRF	5.67	7.23	21.70	-0.242	2.580	1.549	-3.478*
HML	6.81	4.57	14.04	0.405	3.263	2.251	-4.578***
SMB	4.94	5.70	15.40	-0.431	3.322	2.606	-2.870*
WML	6.68	5.00	9.00	0.689	3.698	7.413**	-4.154***
Cyclical Services							
MRF	6.86	8.46	21.24	-0.316	2.799	1.487	-2.985**
HML	9.60	6.72	18.14	0.989	3.422	12.588***	-3.974***
SMB	10.27	9.27	14.67	0.485	4.135	6.390**	-4.167***
WML	3.72	3.57	13.17	-0.358	4.005	4.255	-4.478***
Financials							
MRF	5.67	7.23	21.70	-0.242	2.580	1.549	-3.478*
HML	8.30	7.32	11.24	0.847	4.446	15.366***	-5.649***
SMB	10.23	8.61	16.95	0.651	3.932	7.874**	-4.135***
WML	5.44	4.59	15.56	-1.173	8.242	103.615***	-7.011***
General Industries							
MRF	5.67	7.23	21.70	-0.242	2.580	1.549	-3.478**
HML	10.43	9.56	12.55	1.655	10.270	201.848***	-5.365***
SMB	16.03	13.43	23.69	3.950	25.540	1823.498***	-5.314***
WML	1.24	4.66	22.09	-5.012	33.356	3271.297***	-5.426***
Information Technology							
MRF	1.68	5.77	23.11	-0.456	2.055	2.682	-2.186
HML	28.74	5.49	64.62	3.572	17.314	317.094***	-7.393***
SMB	16.56	14.29	42.89	2.529	12.258	136.635***	-7.430***
WML	-7.59	-2.52	28.68	-2.568	11.418	119.402***	-6.205***
Non-Cyclical Consumer Goods							
MRF	0.86	5.41	22.95	-0.423	2.045	2.492	-2.249
HML	9.26	6.72	26.40	-0.542	3.841	1.986	-3.158**
SMB	18.89	19.16	30.59	-0.008	2.564	0.477	-2.505
WML	4.06	7.43	26.86	0.147	2.791	0.302	-3.742***
Resources							
MRF	10.02	10.20	8.95	-0.774	3.999	5.896*	-0.941
HML	27.02	13.12	42.60	1.152	3.446	10.151***	-3.354**
SMB	64.46	55.23	42.80	1.003	3.974	8.865**	-3.023**
WML	11.72	8.36	44.53	-0.167	3.419	0.365	-1.877
Utilities							
MRF	1.72	5.41	22.76	-0.467	2.118	2.631	-1.976
HML	3.16	1.68	13.03	0.198	2.228	1.384	-2.014
SMB	9.88	9.66	15.77	0.039	1.973	1.896	-1.773
WML	-1.09	-1.24	8.48	0.015	1.993	1.829	-5.609***
Industry							
MRF	5.67	7.23	21.70	-0.242	2.580	1.549	-3.478**
HML	6.52	5.42	9.84	0.401	3.040	2.093	-4.552***
SMB	12.15	12.48	15.01	0.950	6.780	55.814***	-3.013**
WML	3.67	5.14	12.15	-2.401	13.347	413.811***	-4.966***
Service							
MRF	5.67	7.23	21.70	-0.242	2.580	1.549	-3.478**
HML	7.25	7.75	10.66	1.081	5.629	36.147***	-5.018***
SMB	10.45	10.74	13.55	0.613	4.626	12.609***	-3.981***
WML	4.78	5.29	11.53	-0.814	5.105	21.850***	-6.693***

First, albeit the summary statistics for a quarterly and a semi-annual rebalancing reveal a somewhat consistent view on the risk factor as regards magnitude and tendency, it appears that our risk factors become slightly less stationary as turnover frequency decreases. Put differently, we fail to reject the null hypothesis for the presence of unit roots more often in case of a semi-annual or annual portfolio rebalancing than we do for a quarterly portfolio turnover. Yet, given that most of our variables show level stationarity when considering the Augmented Dickey-Fuller (ADF) test statistic (see Said and Dickey, 1984; Dickey and Fuller, 1979), given

a constant and setting the lag p equal to 1, we are confident in obtaining meaningful regression estimates in using an annual frequency.¹⁶

Second, generally one might expect quarterly rebalanced portfolios to show a somewhat superior performance because a more frequent turnover implies the use of more recent data. Thus, when portfolios are rebalanced more frequently, the not entirely lasting explanatory power of the risk factors *HML*, *SMB*, and *WML* may be grasped more effectively.¹⁷ Nonetheless, when taking practical considerations that allow for transaction costs into account, any potential gains associated with a high frequency strategy may be consumed by the costs associated with rebuilding the portfolios. Besides, the book-to-market value that we obtain per month through Datastream always refers to the latest book value shown on the balance sheet, i.e., for European stocks usually a value as of December 31. Thence, it appears more coherent to consider primarily annual rebalanced portfolios. For intra-annual rebalancing frequencies the *HML* factor is inconsistent because it is always based on the book-to-market value at the end of the previous fiscal year.

When looking at Tables 2 and 3, it becomes apparent that about one half of the variables are not normally distributed. This is in line with past empirical findings (see Cochrane, 2005). In about 50% of the cases we reject the null hypothesis of normally distributed data at a 1% significance level when considering the Jarque-Bera test statistic (Jarque and Bera, 1980, 1981).¹⁸ These findings are also underpinned by simply looking at our documented results for

¹⁶The Augmented Dickey-Fuller (ADF) test constructs a parametric correction for higher-order correlation assuming that a variable y follows an autoregressive process $AR(p)$ and adding p lagged difference terms of the dependent variable y to the right hand side of the test regression, i.e.,

$$\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \delta_1 \Delta y_{t-1} + \dots + \delta_p \Delta y_{t-p} + \epsilon_t$$

where α is a constant (here: $\alpha \neq 0$), β the coefficient on a time trend (here: $\beta = 0$) and p the lag order of the AR process (here: $p = 1$). The unit root test is then carried out under the null hypothesis $\gamma = 0$ against the alternative hypothesis of $\gamma < 0$ and evaluated using the test statistic

$$DF \equiv T - ratio = \frac{\hat{\gamma}}{SE(\hat{\gamma})}$$

where $\hat{\gamma}$ is the estimate of γ and $SE(\hat{\gamma})$ is the standard error of the coefficient. If the test statistic is smaller than the critical value for the Dickey-Fuller test, then the null hypothesis of $\gamma = 0$ is rejected, implying that no unit roots are present.

¹⁷For instance, while the book-to-market ratio serves as an extremely good performance predictor of future return for well diversified portfolios (Haugen, 1999), the prospects of stocks alter and assets may change from expensive to cheap and back.

¹⁸The Jarque-Bera test is a goodness-of-fit measure of deviations from normality. It is based on the sample skewness and kurtosis. The test statistic is denoted as

$$JB \equiv \chi^2 - statistic = \frac{N - k}{6} \left(S^2 + \frac{(K - 3)^2}{4} \right) \text{ d.f.} = 2,$$

where N is the number of observations, k represents the number of estimated coefficients, S is the sample skewness, and K is the sample kurtosis. The null hypothesis is a joint hypothesis of $S = 0$ and $K = 3$, since

the third and fourth central moments, i.e., skewness and kurtosis, of the respective variables. *SMB* and *HML* show chiefly a positive skewness, while *MRF* and *WML* appear to be mainly negatively skewed for most countries and industries. In addition, even though most variables possess only somewhat of a small excess kurtosis, quite a few show a kurtosis of 10 or even higher (the highest kurtosis being 33.36 for *WML* for *general industries*).

Intuitively, it appears that the variables for some smaller European economies, such as Portugal, Ireland, Finland, and Sweden, possess a somewhat higher than average kurtosis, which might imply that the returns in these countries are particularly sensitive to market shocks and thus infrequent extreme deviations, such as the "dot-com" bubble. This is supported by high kurtosis values for the *information technology* sector and coinciding positive return fluctuations during the late 1990s and early 2000. Yet, a high kurtosis cannot necessarily be generalized across all smaller economies, as we find rather small kurtosis values for Belgium, Greece, and Denmark, indicating that the variables of these countries show rather modestly-sized deviations. In general, the tests for normality imply that one may want to employ data only after the end of the "dot-com" bubble. This, however, would limit our sample data and make a test for European market integration nearly obsolete, as we expect integration to start only throughout the 1990s. Alternatively, one might include a dummy variable approach in the empirical part in order to correct for the specific event of the "dot-com" bubble.¹⁹

Regardless of the rebalancing frequency considered, our descriptive findings are in line with those of Fama and French (1992, 1993, 1995, 1996, 1998) and Liew and Vassalou (2000) who remark that a value premium is pervasive. For all countries, for the Eurozone, and for all industries, we find that a *HML* portfolio that is long on high book-to-market stocks and short on low book-to-market stocks yields above average market returns. The values range between a median return of 1.00% for the Netherlands and 16.29% for Ireland for the country and Eurozone factors, and between 1.68% for the *utilities* and 13.12% for the *resource* sector when considering industries across the Eurozone.

Furthermore, in line with Banz (1981), Fama and French (1992, 1993, 1995, 1996, 1998), and Liew and Vassalou (2000), we find the existence of a size premium, i.e., the mean and median returns are consistently higher for small firm portfolios relative to big firm portfolios, except for Portugal and *basic industries* where we find small negative median, though positive mean,

samples from a normal distribution have an expected skewness of 0 and an expected kurtosis of 3.

¹⁹Return time plots and histograms allowed us to detect the events associated with the highest fluctuations in returns. Again, due to space considerations, the figures are not presented here. They can, however, be requested from the authors.

returns for *SMB*. This is contrary to the findings of Otten and Bams (2002) who document the existence of a growth effect (i.e., big stocks outperform small stocks) in major European markets. One explanation for our discrepancy with the findings of Otten and Bams (2002) might be that our sample period differs from that of the former who focus exclusively on the period 1991 to 1998, and thus, ex-ante the "dot-com" bubble. Overall, Table 2 reveals that the median returns for *SMB* vary between -1.80% for Portugal (yet, mean return of 4.41%) and 13.73% for Finland (the mean return is here even 28.65%). The deviations are even higher for the industries. For instance, Table 3 reflects median returns between -1.47% for *basic industries* and 55.23% for the *resource* sector across the Eurozone.

Regarding the profitability of a momentum strategy, our results support the findings of Jegadeesh and Titman (1993), Liew and Vassalou (2000), and Rouwenhorst (1998). It appears that past winner stocks usually outperform past loser stocks in the short run for nearly all countries, except Portugal, Denmark, and Sweden, all industries (except the *information technologies* and *utilities* sector), and across the Eurozone as a whole. For momentum, the rebalancing period, i.e., annually, semi-annually, or quarterly, makes, however, a considerable difference.²⁰ It seems as if momentum is very sensitive to the rebalancing frequency. The more often the portfolios are rebalanced, the higher become (on average) their mean and median returns. In other words, returns to *WML* decrease significantly as the turnover interval increases. Yet, any potential financial gains associated with a higher turnover may eventually offset by higher transaction costs.

From an investor's perspective it is worthy to note that on average the standard deviations for *WML* tend to be smaller than those of *HML*, and *SMB*. This implies a less risky investment opportunity. In addition, the highest *WML* median return that we found for a country was merely 6.25% for Belgium, a value which is only about two third of the country's *SMB* return of 9.42%. The difference becomes even higher when we consider industries. Here, we find the highest *WML* median return for the *resource* sector with a value of 8.36%. Albeit this seems to be a decent gain above an average market return, it looks rather small when compared to the previously mentioned *SMB* median return of 55.23% for the same sector.

Overall, our findings support the existence of a value, size, and momentum effect, not only for different countries, but also for different industries, and even across the Eurozone. Especially, the latter might be of interest to an investor who simultaneously (i) wants to hold a diversified portfolio in line with modern portfolio theory as firstly proposed by Markowitz

²⁰Note that the results are not reported here. They may, however, be requested from the authors.

(1952) and yet (ii) intends to surpass the market by following an investment strategy driven by a value, size, or momentum effect. By investing in a pan-Eurozone portfolio, an investor may not put all his eggs in one basket, i.e., one country or industry, but still take advantage of any present anomalies.

4.4 Returns on *HML*, *SMB*, and *WML* at Different States of the Economy

Like Liew and Vassalou (2000) we study the returns on our factors, i.e., the *HML*, *SMB*, and *WML* trading strategies, at different states of the economy. We use quarterly observations and associate next year's annual *GDP* growth with past year's annual return per trading strategy.²¹ We sort by growth in *GDP* every quarter. For each country we use the continuously compounded growth rate in the respective country's seasonally adjusted Gross Domestic Product. For the Eurozone, as well as for each industry, the *GDP* growth rate is calculated as the continuously compounded rate in the Eurozone's Gross Domestic Product, which is as well seasonally adjusted.

We define as 'good states' of the economy those states that exhibit the highest 33.33% future *GDP* growth rate per country or in the Eurozone. 'Bad states' are those states that exhibit the lowest 33.33% future *GDP* growth. The remaining third is classified as 'mid state'.²² The presented Δ depicts the difference between the 'good states' and the 'bad states' of the respective economies. *T*-values are computed for this difference.²³ The findings per country are depicted in Table 4. Results per industry are shown in Table 5.

The results indicate that *SMB* is primarily positively related to future growth in *GDP* for both countries (in 13 out of 17 cases) and industries (in 9 out of 11 cases). The difference between 'good' and 'bad' states of the economy is statistically significant for most of the countries and industries under consideration. In other words, small stocks are striving prior to an economic recovery and boom, given that the returns in projection of 'bad states' are on average lower than those for the 'good states'. This is in line with the findings of Liew and Vassalou (2000). A clear exception to this constitute our findings for the *resources* sector where we find actually an inverse relationship between returns to *SMB* and future economic growth. Although we would in fact expect a strong positive relation for this sector, given that

²¹Please note that we employ annually rebalanced portfolios to obtain our risk factors; please refer to section 4.2 for details.

²²In this respect our study differs from that of Liew and Vassalou (2000) as the latter do not include a mediocre state and classify 'good' ('bad') states as those that exhibit the highest (lowest) 25% of future *GDP* growth.

²³The *T*-statistic is computed by dividing the difference between the returns on the 'good states' and 'bad states' by the quotient of the standard deviation of the returns over the square root of the number of observations. More formally: $\frac{[R_{GS} - R_{BS}]}{\sigma / \sqrt{n}}$.

Table 4: Performance of Risk Factors at Different States of the Economy per Country - 4 Quarter Lag

The results are based on annually rebalanced HML, SMB, and WML portfolios using quarterly observations. HML is the annual return on a portfolio that is long on high book-to-market stocks and short on low book-to-market securities, holding size and momentum characteristics of the portfolio constant. SMB is the annual return on a portfolio that is long on small capitalization stocks and short on big capitalization securities, holding book-to-market and momentum characteristics of the portfolio constant. WML is the annual return on a portfolio that is long on the best performing stocks of the past year ('winners') and short on the worst performing securities of the previous year ('losers') holding book-to-market and size characteristics of the portfolio constant. The GDP growth rate is calculated as the continuously compounded rate in a country's Gross Domestic Product, which is seasonally adjusted. We define as 'good states' of the economy those states that exhibit the highest 33.33% future GDP growth rate in the individual countries/the Eurozone. 'Bad states' are those states that exhibit the lowest 33.33% future GDP growth. The remaining third is classified as 'mid state'. The presented Δ depicts the difference between the 'good states' and the 'bad states' of the respective economies. T -values are computed for this difference.

Country	Past year return on factor sorted by future GDP growth														
	HML					SMB					WML				
	Good State (%)	Mid State (%)	Bad State (%)	Δ Go./Bad (%)	T-value	Good State (%)	Mid State (%)	Bad State (%)	Δ Go./Bad (%)	T-value	Good State (%)	Mid State (%)	Bad State (%)	Δ Go./Bad (%)	T-value
Austria	19.89	19.90	2.34	17.55	3.54	9.90	21.39	5.02	4.88	1.09	4.48	5.86	1.38	3.10	1.21
Belgium	1.39	10.03	5.35	-3.95	-2.20	15.23	4.71	5.74	9.49	4.79	14.21	2.86	-1.00	15.21	7.13
Finland	20.52	38.88	21.98	-1.46	-0.15	32.46	46.19	11.92	20.54	2.04	0.94	2.01	-5.92	6.86	4.26
France	6.21	5.77	5.47	0.74	0.45	10.90	9.46	5.57	5.32	2.81	3.77	1.82	3.80	-0.03	-0.03
Germany	1.56	6.67	18.09	-16.53	-7.37	5.75	6.35	19.00	-13.25	-4.27	2.85	6.44	5.36	-2.51	-2.30
Greece	4.63	15.19	-6.60	11.24	2.27	4.45	2.25	-14.08	18.53	5.20	-6.01	-6.78	6.78	-12.79	-3.06
Ireland	8.95	46.15	29.95	-21.00	-3.05	7.45	23.69	-10.74	18.19	2.60	-8.59	-10.06	-1.83	-6.76	-1.12
Italy	7.55	-3.72	9.29	-1.74	-1.09	8.93	7.32	0.85	8.07	4.19	1.86	3.81	3.58	-1.73	-1.32
Netherlands	1.71	4.23	9.88	-8.17	-4.12	9.35	11.00	-1.13	10.48	4.97	3.39	6.29	-1.23	4.62	2.99
Portugal	22.55	15.79	6.13	16.41	2.67	19.04	13.84	-2.25	21.29	2.95	-10.21	-5.25	-1.28	-8.94	-3.33
Spain	9.69	15.32	8.72	0.97	0.38	13.52	10.96	17.04	-3.52	-0.86	-1.91	4.14	-9.75	7.84	2.62
Denmark	10.42	7.83	29.40	-18.98	-5.79	22.87	22.31	6.17	16.69	3.80	-1.52	2.20	0.22	-1.74	-0.60
Sweden	25.69	7.59	6.30	19.40	3.51	3.47	10.33	12.07	-8.60	-2.76	-12.63	4.12	-4.76	-7.88	-2.57
United Kingdom	6.52	6.43	3.50	3.03	2.64	5.89	7.58	14.62	-8.73	-4.67	3.49	3.83	-3.25	6.74	5.46
Norway	9.84	7.33	-0.71	10.55	4.75	6.26	0.32	2.56	3.70	1.64	0.38	3.10	6.04	-5.67	-2.59
Switzerland	17.73	24.93	3.51	14.23	3.19	13.85	19.24	6.09	7.75	1.98	3.70	-12.25	-2.07	5.77	1.92
Eurozone	6.81	7.17	6.36	0.45	0.45	19.37	9.48	3.20	16.17	10.94	3.73	5.16	2.73	0.99	0.92

Table 5: Performance of Risk Factors at Different States of the Economy per Industry (annually rebalanced)

The results are based on annually rebalanced HML, SMB, and WML portfolios using quarterly observations. HML is the annual return on a portfolio that is long on high book-to-market stocks and short on low book-to-market securities, holding size and momentum characteristics of the portfolio constant. SMB is the annual return on a portfolio that is long on small capitalization stocks and short on big capitalization securities, holding book-to-market and momentum characteristics of the portfolio constant. WML is the annual return on a portfolio that is long on the best performing stocks of the past year ('winners') and short on the worst performing securities of the previous year ('losers') holding book-to-market and size characteristics of the portfolio constant. The GDP growth rate is calculated as the continuously compounded rate in the Eurozone's Gross Domestic Product, which is seasonally adjusted. We define as 'good states' of the economy those states that exhibit the highest 33.33% future GDP growth rate in the individual industries. 'Bad states' are those states that exhibit the lowest 33.33% future GDP growth. The remaining third is classified as 'mid state'. The presented Δ depicts the difference between the 'good states' and the 'bad states' of the respective economies. T -values are computed for this difference.

Industry	Past year return on factor sorted by future GDP growth														
	HML					SMB					WML				
	Good State (%)	Mid State (%)	Bad State (%)	Δ Go./Bad (%)	T-value	Good State (%)	Mid State (%)	Bad State (%)	Δ Go./Bad (%)	T-value	Good State (%)	Mid State (%)	Bad State (%)	Δ Go./Bad (%)	T-value
Basic Industries	19.86	7.44	5.55	14.31	6.18	-2.48	5.30	-1.06	-1.42	-0.54	-1.31	2.54	-3.06	1.75	0.99
Cyclical Consumer Goods	7.46	7.73	8.02	-0.56	-0.37	10.81	6.69	-3.08	13.89	7.90	4.17	6.12	1.69	2.48	2.43
Cyclical Services	5.88	7.08	6.02	-0.14	-0.08	14.59	14.54	10.57	4.03	2.16	6.47	-1.04	0.61	5.86	3.82
Financials	10.24	8.87	7.49	2.75	1.82	14.21	6.86	2.59	11.63	8.25	-2.31	5.32	3.12	-5.43	-4.13
General Industries	15.93	9.59	5.22	10.71	7.05	22.46	8.70	7.91	14.55	7.02	-0.92	4.72	1.35	-2.27	-1.09
Information Technology	25.88	33.99	15.80	10.08	1.22	22.50	21.71	3.59	18.91	4.25	-7.94	-13.92	-7.40	-0.54	-0.12
Non-Cyclical Consumer Goods	-2.00	20.44	23.56	-25.57	-6.94	29.27	4.18	6.42	22.84	4.20	9.88	1.04	6.39	3.49	0.69
Resources	38.66	25.34	61.08	-22.42	-1.37	35.75	59.27	158.38	-122.63	-5.51	24.65	36.70	-21.53	46.18	2.65
Utilities	-3.61	3.80	8.82	-12.43	-4.87	21.66	9.00	10.57	11.09	3.50	-3.53	2.08	5.39	-8.92	-4.16
Industry	7.98	6.53	5.18	2.80	2.32	15.98	10.42	3.40	12.58	7.90	3.01	3.59	2.41	0.59	0.49
Service	8.70	6.61	3.56	5.14	4.36	14.14	8.33	3.56	10.59	7.80	-0.91	3.76	2.12	-3.04	-2.63

more resources are demanded whenever an economy is facing an upward trend, the negative relation reported might be explained by our very small time frame for this particular sector, spanning only from April 2004 to April 2007.²⁴

Surprisingly are also our findings for Germany and the inverse relation between *SMB* and future GDP growth. This might be explained by the belief that smaller firms may react faster and more efficient to future market changes, given their higher flexibility when compared to bigger firms. As bigger and more diversified companies may follow more the general market trend, and may thus also be more punished during economic down-swings, smaller firms may follow a niche and may thus be more immune to general real economic activities. Yet, whether this seems to be the case in Germany is purely speculative. Alternatively, the non-normal behavior for the German GDP growth rate might serve as an explanation.

In general, our findings for *SMB* underpin Perez-Quiros and Timmermann (2000), who remark that the returns on small firms are more volatile during economic recessions than peaks, given the increased sensitivity of investors towards risk. In other words, small firms appear to be extremely sensitive to economic swings, due to e.g., liquidity constraints and a lack of diversification. This sensitivity is especially reflected in the findings depicted in Table 5, which highlight the very sensitive, but positive, relation between the return to *SMB* of the respective industries and future growth in GDP. Finally, as we find, with the exception of Greece, Ireland, the Netherlands, and Portugal, as well as *basic industries* and *cyclical consumer goods*, positive returns to *SMB* for all stages of the economy, we may consider this a support for the existence of a robust size effect.

As regards *HML*, a robust positive relation between the factor returns and future GDP growth cannot be as easily identified. Although 16 out of 28 cases show a positive and mostly significant difference between the factor returns during 'good' and 'bad' states of the economy, there is not a very clear pattern. Our findings are thus not entirely in line with those of Liew and Vassalou (2000), which may, however, be explained by different samples as regards both, countries/industries as well as the time frame under consideration and the slightly diverging classification of 'good' and 'bad' states of the economy. Overall, however, we find some, though not necessarily very strong, support for Lettau and Ludvigson (2001), who suggest that *HML* is particularly sensitive to bad news in bad times. In those cases, were we obtain a negative relation, our results imply that low book-to-market and big capitalization firms are actually

²⁴Albeit Figure 4 depicts that our *resources* samples spans from April 2004 to April 2008, the last year becomes obsolete, due to a four quarter time lag.

better able to thrive than high book-to-market and small capitalization stocks when strong economic growth is expected. Thence, investors would rather hold stocks whose returns are relatively high when they realize that the economy is in anticipation of a recovery or boom phase. Although these results may seem to be a bit odd at first glance, one plausible explanation for the relatively stronger performance of high book-to-market stocks during lower economic growth phases may lay in a more fundamental, solid, and conservative investment approach by investors. Even if high book-to-market firms are often seen as those companies that possess less growth opportunities, they may alternatively be considered those that are traded closer to their fundamental values, and, hence, less subject to psychological market influences. Thus, during negative economic trends, more pessimistic investors may be inclined to invest in high book-to-market stocks as opposed to low-book-to-market ones.

Finally, the results depicted for *WML* in Table 4 and 5 do not provide any clear pattern for the relationship between these factor returns and future GDP growth. In fact, we have a tie between positive and negative relations. One explanation for the unclear pattern of *WML* with respect to future GDP growth might be that investors tend to mistakenly project a continuation of abnormal profit levels long periods into the future. This is not necessarily in accordance with real economic activity. In consequence, successful firms become overvalued and unsuccessful ones undervalued vis-à-vis the firms' fundamentals. This implies that the market reacts inefficiently. The market develops a belief that a few positive or negative events cause a run that will persist for long periods into the future. Haugen (1999), for instance, argues that if a firm's earnings per share have increased quickly over the last quarters, the market assumes that this foretells continuous success for many more quarters into the future. Therefore, the price of the share becomes inflated on the basis of this expectation. The market is wrong as past success does not foretell prolonged future success. Due to this inefficiency, there may be no clear traceable pattern between *WML* and future real economic activity.

Overall, the information contained in the returns to *HML*, *SMB*, and to a lesser extent to *WML* about future real economic activities is largely independent of the information contained in the market factor. The results from our multicollinearity analysis (cf. Tables I and II in the Appendix) show that there exist no linear relation between the returns to the individual risk factors and the market. Thence, the relation between the risk factors and future economic growth is unlikely to be induced by the leading relation between the market factor and real future economic activities as suggested in a variety of past studies (see Fischer and Merton, 1984; Aylward and Glen, 2000; Fama, 1981, 1990; Geske and Roll, 1983; Barro, 1990; Schwert,

1990; Mullins and Wadhwani, 1989; Wahlroos and Berglund, 1986; Wasserfallen, 1989, 1990; Binswanger, 2000a,b, 2004; Liew and Vassalou, 2000). This hypothesis is confirmed by the results of the following section.

5 Regression Results and Implications

The following paragraphs discuss our regression results that reflect the relation between the various trading strategies, i.e., *HML*, *SMB*, and *WML*, and future growth in *GDP* for various countries and industries in the Eurozone. For reasons mentioned above, we focus our analysis on annually rebalanced portfolios for the construction of the risk factors. The parameters depicted are corrected for heteroscedasticity and autocorrelation, up to three lags, using the Newey and West (1987) estimator.

5.1 One-Factor Regressions

In regressing future GDP growth on market excess returns as pointed out in equation 2, we find, in line with Fama (1981) and Liew and Vassalou (2000) primarily a positive relation between these variables. This holds for both European countries and industries aggregated across the Eurozone.

In particular, Table 6 reveals that a regression of future economic growth on past excess market returns results mainly in positive and partly statistically significant coefficients. Yet, given the fractionally negative adjusted R^2 figures, it appears that the market factor alone does not necessarily play a good role in explaining future economic development.²⁵ The results are confirmed when shifting the view from countries to industries. The results for the latter are depicted in Table 7. Again, the *resources* sector takes on an inverse role, which, however, may once more be explained by the small sample size. As such the findings should be treated with caution. In addition, it is worthy to note that some of the industries reveal the same statistics for the market factor. This is due to the fact that in the industry regressions the dependent variable always constitutes the GDP growth rate of the Eurozone and that the market factor is always represented by the excess return to the DJ Euro index. Moreover the sample pe-

²⁵Unlike R^2 , which gives an indication about how much the ability to predict is improved by using the regression line (as opposed to not using it), the adjusted R^2 may take a value lower than 0. An adjusted R^2 tries to take into account that an explanatory variable X (here: *MRF*, *HML*, *SMB*, and *WML*) that is supposed to be completely unrelated to the dependent variable Y (here: GDP_{growth}) will probably have some relationship to Y . The adjusted R^2 reduces the R^2 by how much fit would probably happen just by luck. In some cases this reduction results in a negative adjusted R^2 . For the smaller economies, a negative adjusted R^2 can also be explained by the fact that the ratio of observations to regressors is too low, given the small sample size

Table 6: Univariate Regressions of GDP Growth Conditional on Past Factor Returns per Country - 4 Quarter Lag

$$GDP_{growth(t,t+4)} = \alpha + \beta Factor Ret_{t-4,t} + \epsilon_{t,t+4}$$

In the regression notation, *Factor Ret* refers to MRF, HML, SMB, and WML. MRF is the market risk premium in each country/the Eurozone. The risk free rate is given by the one-month ecu deposit quoted in London. The regressions use annually rebalanced HML, SMB, and WML portfolios. HML is the annual return on a portfolio that is long on high book-to-market stocks and short on low book-to-market securities, holding size and momentum characteristics of the portfolio constant. SMB is the annual return on a portfolio that is long on small capitalization stocks and short on big capitalization securities, holding book-to-market and momentum characteristics of the portfolio constant. WML is the annual return on a portfolio that is long on the best performing stocks of the past year ('winners') and short on the worst performing securities of the previous year ('losers') holding book-to-market and size characteristics of the portfolio constant. GDP is calculated as the continuously compounded growth rate in each country/the Eurozone. The GDP is seasonally adjusted. *T*-statistics are corrected for heteroscedasticity and autocorrelation, up to three lags, using the Newey and West (1987) estimator. The adjusted R^2 is corrected for degrees of freedom. *, **, and *** are used as indicators of statistical significance at, respectively, the 10 %, 5%, and 1% significance level.

Country	Slope Coefficients				<i>T</i> -Statistics				Adjusted R^2 (%)			
	MRF	HML	SMB	WML	MRF	HML	SMB	WML	MRF	HML	SMB	WML
Austria	0.044	0.023	0.013	0.002	6.510***	1.703*	0.735	0.070	66.637	11.761	-1.383	-6.214
Belgium	0.015	-0.014	0.031	0.039	1.303	-0.483	1.656*	2.605***	3.828	-0.988	6.089	13.411
Finland	0.030	0.005	0.010	0.063	4.110***	1.667*	1.518	1.476	40.125	-1.598	3.575	3.287
France	0.011	0.009	0.024	-0.005	0.922	0.481	1.370	-0.322	1.491	-0.791	4.620	-1.341
Germany	-0.004	-0.035	-0.014	-0.053	-0.240	-2.929***	-2.015**	-1.398	-1.409	11.906	2.557	5.786
Greece	-0.005	0.008	0.021	-0.015	-1.203	0.988	2.871***	-2.381**	-2.067	-0.194	14.387	8.397
Ireland	0.016	-0.015	0.016	-0.019	0.464	-0.879	1.477	-1.336	-3.326	-1.133	-0.290	-0.122
Italy	-0.007	0.011	0.038	-0.022	-0.454	0.323	1.476	-0.726	-0.891	-1.130	4.389	-0.540
Netherlands	0.036	-0.022	0.016	0.004	3.278***	-1.348	1.258	0.188	19.869	3.401	1.512	-1.342
Portugal	-0.004	0.010	0.010	-0.039	-0.168	1.391	2.255**	-1.982**	-3.775	0.379	2.182	8.176
Spain	0.006	-0.006	0.004	0.001	1.080	-0.701	0.922	0.147	-0.085	-1.279	-0.655	-2.322
Denmark	0.008	-0.030	0.021	-0.006	0.438	-1.997	1.523	-0.428	-1.933	7.408	5.737	-2.832
Sweden	0.014	0.019	-0.019	-0.026	1.409	3.884***	-1.220	-1.725*	7.101	18.128	4.259	9.310
United Kingdom	-0.011	0.024	-0.011	0.017	-0.705	1.393	-1.198	1.739*	-0.341	1.090	-0.024	-0.060
Norway	0.000	0.005	0.037	-0.043	0.001	0.121	0.916	-1.336	-1.429	-1.394	0.348	0.879
Switzerland	-0.005	0.004	0.003	0.014	-0.353	0.403	0.284	1.418	-1.615	-1.437	-1.800	1.717
Eurozone	0.009	0.005	0.063	0.006	0.746	0.232	3.181***	0.238	0.038	-1.354	22.017	-1.297

riod coincides for these particular industries, i.e., *cyclical consumer goods*, *financials*, *general industries*, *industry*, and *service*. Again, we choose the GDP growth rate of the Eurozone for all industries as we want to test whether there are common state variables across European industries that may proxy for future pan-Eurozone economic activities. If this is the case, then this may provide empirical support for the integration of equity markets across the Eurozone.

Perhaps more interesting is the fact that we show a positive and equally strong relation for *SMB* and future economic growth. Yet, the countries and industries are not necessarily the same as for which we find the strongest positive relation of the market factor and future growth in GDP. This implies that whenever the market factor itself may not contain information about future real activities, then there might be a chance that the return to *SMB* may provide such information. The explanatory power of *SMB* per industry seems to be somewhat higher than that per country. Albeit this might come a bit surprising, it further adds to the evidence of stock market integration and underpins our hypothesis of section 4.4.

The regression results for *HML* and *WML* depicted in Tables 6 and 7 are also in line with our previously discussed findings. There is a small tendency for a positive relation between *HML* and future GDP growth, for both countries and industries, and a somewhat mixed, i.e., partly negative and partly positive, relation between *WML* and future macroeconomic growth. This is in line with Liew and Vassalou (2000). As for *HML*, a positive relation implies that high book-to-market stocks are better able to prosper than low book-to-market stocks whenever periods of high economic growth are anticipated. Liew and Vassalou (2000) suggest that a positive relation makes sense as investors prefer to hold stocks whose returns are relatively high when they realize that the economy is in a bad state. They thence prefer and tend to hold low book-to-market stocks with good growth opportunities and typically low debt ratios.

Concerning *WML*, a straightforward explanation does not come as obvious. The partly negative and partly positive relation to GDP may perhaps be treated on a case by case basis, presuming the presented results are robust, i.e., irrespective of the sample time period chosen. Alternatively, one may argue that on average *WML* may contain little, if any, information regarding future economic development.

Overall, albeit some minor lack of robustness, our findings may allow us to make some indicative and consistent inference from past factor returns towards future real economic activity. This holds not only for individual countries, but also for different industries aggregated across the Eurozone. Under market integration consideration, the latter is of particular interest since it provides not only some empirical support for the fact that the Fama and French

Table 7: Univariate Regressions of GDP Growth Conditional on Past Factor Returns per Industry - 4 Quarter Lag

$$GDP_{growth(t,t+4)} = \alpha + \beta FactorRet_{t-4,t} + \epsilon_{t,t+4}$$

In the regression notation, *FactorRet* refers to MRF, HML, SMB, and WML. MRF is the market risk premium in each industry. The risk free rate is given by the one-month eu deposit quoted in London. The regressions use annually rebalanced HML, SMB, and WML portfolios. HML is the annual return on a portfolio that is long on high book-to-market stocks and short on low book-to-market securities, holding size and momentum characteristics of the portfolio constant. SMB is the annual return on a portfolio that is long on small capitalization stocks and short on big capitalization securities, holding book-to-market and momentum characteristics of the portfolio constant. WML is the annual return on a portfolio that is long on the best performing stocks of the past year ('winners') and short on the worst performing securities of the previous year ('losers') holding book-to-market and size characteristics of the portfolio constant. GDP is calculated as the continuously compounded growth rate in each industry. The GDP is seasonally adjusted. *T*-statistics are corrected for heteroscedasticity and autocorrelation, up to three lags, using the Newey and West (1987) estimator. The adjusted R^2 is corrected for degrees of freedom. *, **, and *** are used as indicators of statistical significance at, respectively, the 10 %, 5%, and 1% significance level.

Country	Slope Coefficients				<i>T</i> -Statistics				Adjusted R^2 (%)			
	MRF	HML	SMB	WML	MRF	HML	SMB	WML	MRF	HML	SMB	WML
Basic Industries	0.013	0.009	0.003	0.008	1.950*	0.838	0.168	0.745	4.792	0.413	-1.430	-0.653
Cyclical Consumer Goods	0.008	-0.007	0.032	0.014	0.760	-0.529	1.756*	0.545	0.076	-1.099	7.783	-0.845
Cyclical Services	0.019	0.015	0.000	0.020	2.198**	0.658	0.039	1.443	7.038	0.605	-1.490	1.725
Financials	0.008	0.005	0.039	-0.016	0.760	0.373	2.329**	-1.361	0.076	-1.243	7.271	-0.123
General Industries	0.008	0.048	0.029	-0.008	0.760	2.535**	2.256**	-0.878	0.076	14.552	9.266	-0.512
Information Technology	0.019	0.001	0.010	-0.007	3.407***	0.172	1.821*	-0.717	39.380	-4.245	4.681	-0.547
Non-Cyclical Consumer Goods	0.017	-0.022	0.006	0.001	3.200***	-2.733***	1.740*	0.220	35.699	29.311	0.580	-4.350
Resources	-0.014	-0.003	-0.009	0.006	-0.626	-0.599	-22.909***	1.979**	-15.422	-15.154	92.114	12.479
Utilities	0.020	-0.020	0.010	-0.031	3.290***	-2.231**	1.233	-2.372**	31.082	5.635	-0.230	12.010
Industry Service	0.008	0.020	0.040	0.000	0.760	0.896	2.054**	0.012	0.076	0.338	10.357	-1.428
	0.008	0.029	0.029	-0.003	0.760	1.360	1.643	-0.231	0.076	1.936	3.156	-1.390

(1993) factors may serve as state variables within the context of Merton's ICAPM (Merton, 1973), but also that equity markets are integrated to a certain degree. This holds in particular when focusing on the equity factor returns in some particular industries, such as the *resources*, *cyclical services*, and *cyclical consumer goods* sector, given that these contain considerable incremental information on future economic development in the Eurozone.

5.2 Two-Factor Regressions

In this section we compare the information content of *HML*, *SMB*, and *WML* with that of *MRF* to determine whether any of the additional three risk factors comprises incremental explanatory power relative to the market factor on the future state of the economy. In particular, we regress future GDP growth simultaneously on the market factor and one of the three risk factors (cf. equation 3). Table 8 reports the results for annually rebalanced portfolios per country, while Table 9 depicts the findings for different industries aggregated across the Eurozone.

Table 8: Bivariate Regressions of GDP Growth Conditional on Past Factor Returns per Country - 4 Quarter Lag

$GDP_{growth(t,t+4)} = \alpha + \beta MRF_{t-4,t} + \gamma FactorRet_{t-4,t} + \epsilon_{t,t+4}$					
In the regression notation, <i>FactorRet</i> refers to HML, SMB, and WML. MRF is the market risk premium in each country/the Eurozone. The risk free rate is given by the one-month ecu deposit quoted in London. The regressions use annually rebalanced HML, SMB, and WML portfolios. HML is the annual return on a portfolio that is long on high book-to-market stocks and short on low book-to-market securities, holding size and momentum characteristics of the portfolio constant. SMB is the annual return on a portfolio that is long on small capitalization stocks and short on big capitalization securities, holding book-to-market and momentum characteristics of the portfolio constant. WML is the annual return on a portfolio that is long on the best performing stocks of the past year ('winners') and short on the worst performing securities of the previous year ('losers') holding book-to-market and size characteristics of the portfolio constant. GDP is calculated as the continuously compounded growth rate in each country/the Eurozone. The GDP is seasonally adjusted. <i>T</i>-statistics are corrected for heteroscedasticity and autocorrelation, up to three lags, using the Newey and West (1987) estimator. The adjusted R^2 is corrected for degrees of freedom. *, **, and *** are used as indicators of statistical significance at, respectively, the 10 %, 5%, and 1% significance level.					
Country	MRF		HML		Adj. R^2 (%)
	Slope	T-Statistics	Slope	T-Statistics	
Austria	0.043	6.412***	0.020	3.656***	79.596
Belgium	0.015	1.224	-0.012	-0.428	2.481
Finland	0.039	5.768***	-0.015	-2.736***	47.013
France	0.011	0.946	0.008	0.525	0.592
Germany	0.003	0.222	-0.036	-3.384***	10.567
Greece	-0.010	-1.657*	0.014	1.681*	6.038
Ireland	0.029	0.622	-0.019	-0.930	-2.914
Italy	-0.013	-0.782	0.022	0.713	-1.272
Netherlands	0.035	3.145***	-0.017	-1.393	21.628
Portugal	-0.010	-0.468	0.012	2.393**	-2.327
Spain	0.005	0.852	-0.004	-0.468	-1.987
Denmark	0.013	0.815	-0.034	-2.285**	7.692
Sweden	0.005	0.601	0.017	2.651***	17.271
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Table 8 – continued from previous page

United Kingdom	-0.012	-0.808	0.026	1.520	1.131
Norway	-0.003	-0.084	0.007	0.164	-2.849
Switzerland	-0.007	-0.511	0.005	0.567	-2.694
Eurozone	0.009	0.758	0.007	0.328	-1.270

Country	MRF		SMB		Adj. R^2 (%)
	Slope	T-Statistics	Slope	T-Statistics	
Austria	0.044	6.652***	0.014	1.922*	70.119
Belgium	0.009	0.704	0.024	1.112	5.722
Finland	0.042	7.068***	-0.015	-3.035***	46.192
France	0.009	0.819	0.022	1.352	5.305
Germany	-0.002	-0.141	-0.014	-1.957*	0.953
Greece	-0.011	-2.715***	0.030	6.114***	27.467
Ireland	0.006	0.174	0.015	1.273	-4.542
Italy	-0.006	-0.378	0.037	1.453	3.324
Netherlands	0.035	3.243***	0.010	1.011	19.926
Portugal	-0.008	-0.413	0.011	3.774***	-0.744
Spain	0.004	0.433	0.002	0.238	-2.272
Denmark	-0.001	-0.046	0.021	1.672*	2.706
Sweden	0.012	1.415	-0.014	-1.192	8.831
United Kingdom	-0.009	-0.570	-0.010	-1.024	-0.765
Norway	0.006	0.240	0.039	1.041	-0.979
Switzerland	-0.007	-0.539	0.004	0.493	-3.151
Eurozone	-0.003	-0.240	0.065	3.085***	21.019

Country	MRF		WML		Adj. R^2 (%)
	Slope	T-Statistics	Slope	T-Statistics	
Austria	0.045	6.656***	-0.014	-1.953*	66.393
Belgium	0.005	0.408	0.035	1.986**	11.881
Finland	0.029	3.888***	0.016	0.395	38.684
France	0.012	0.983	-0.009	-0.689	0.369
Germany	0.000	0.022	-0.054	-1.505	4.135
Greece	-0.005	-1.070	-0.016	-2.630***	7.133
Ireland	0.018	0.613	-0.020	-1.272	-3.396
Italy	-0.006	-0.355	-0.020	-0.610	-1.665
Netherlands	0.037	3.282***	-0.007	-0.310	18.970
Portugal	-0.017	-0.816	-0.048	-2.405**	8.476
Spain	0.007	1.060	0.004	0.461	-1.907
Denmark	0.007	0.404	-0.003	-0.245	-5.148
Sweden	0.009	1.126	-0.019	-1.375	10.421
United Kingdom	-0.010	-0.682	0.016	1.764*	-0.484
Norway	0.008	0.300	-0.046	-1.666*	-0.375
Switzerland	-0.001	-0.102	0.014	1.300	-0.253
Eurozone	0.009	0.725	0.005	0.186	-1.341

Table 9: Bivariate Regressions of GDP Growth Conditional on Past Factor Returns per Industry - 4 Quarter Lag

$$GDP_{growth(t,t+4)} = \alpha + \beta MRF_{t-4,t} + \gamma FactorRet_{t-4,t} + \epsilon_{t,t+4}$$

In the regression notation, *FactorRet* refers to HML, SMB, and WML. MRF is the market risk premium in each industry. The risk free rate is given by the one-month ecu deposit quoted in London. The regressions use annually rebalanced HML, SMB, and WML portfolios. HML is the annual return on a portfolio that is long on high book-to-market stocks and short on low book-to-market securities, holding size and momentum characteristics of the portfolio constant. SMB is the annual return on a portfolio that is long on small capitalization stocks and short on big capitalization securities, holding book-to-market and momentum characteristics of the portfolio constant. WML is the annual return on a portfolio that is long on the best performing stocks of the past year ('winners') and short on the worst performing securities of the previous year ('losers') holding book-to-market and size characteristics of the portfolio constant. GDP is calculated as the continuously compounded growth rate in each industry. The GDP is seasonally adjusted. *T*-statistics are corrected for heteroscedasticity and autocorrelation, up to three lags, using the Newey and West (1987) estimator. The adjusted R^2 is corrected for degrees of freedom. *, **, and *** are used as indicators of statistical significance at, respectively, the 10 %, 5%, and 1% significance level.

Country	MRF		HML		Adj. R^2 (%)
	Slope	T-Statistics	Slope	T-Statistics	
Basic Industries	0.012	1.658*	0.005	0.423	3.696
Cyclical Consumer Goods	0.010	0.860	-0.010	-0.824	-0.680
Cyclical Services	0.018	2.296**	0.010	0.492	6.695
Financials	0.008	0.708	0.004	0.248	-1.281
General Industries	0.004	0.396	0.047	2.434**	13.685
Information Technology	0.022	3.927***	-0.005	-2.137**	43.425
Non-Cyclical Consumer Goods	0.012	1.739*	-0.011	-1.214	38.563
Resources	-0.010	-0.515	-0.002	-0.463	-41.465
Utilities	0.018	3.177***	-0.013	-2.249**	32.210
Industry	0.009	0.820	0.022	1.003	0.732
Service	0.007	0.654	0.027	1.218	1.635

Country	MRF		SMB		Adj. R^2 (%)
	Slope	T-Statistics	Slope	T-Statistics	
Basic Industries	0.013	2.009**	-0.002	-0.111	3.303
Cyclical Consumer Goods	0.004	0.366	0.030	1.593	6.811
Cyclical Services	0.020	2.191**	-0.007	-0.611	6.187
Financials	0.006	0.599	0.037	2.201**	6.760
General Industries	0.003	0.240	0.028	2.042**	8.095
Information Technology	0.025	3.209***	-0.009	-1.725*	40.986
Non-Cyclical Consumer Goods	0.017	2.990***	0.000	0.155	32.674
Resources	-0.006	-2.307**	-0.009	-21.894***	91.327
Utilities	0.019	3.249***	0.006	0.980	29.608
Industry	-0.002	-0.137	0.041	1.780*	9.112
Service	0.007	0.684	0.028	1.497	2.891

Country	MRF		WML		Adj. R^2 (%)
	Slope	T-Statistics	Slope	T-Statistics	
Basic Industries	0.017	2.892***	0.019	1.770*	7.565
Cyclical Consumer Goods	0.008	0.693	0.009	0.393	-1.108
Cyclical Services	0.018	2.103**	0.018	1.466	8.292
Financials	0.009	0.835	-0.017	-1.520	0.096
General Industries	0.008	0.744	-0.008	-0.868	-0.538

Continued on next page

Table 9 – continued from previous page

Information Technology	0.019	3.764***	-0.002	-0.313	37.136
Non-Cyclical Consumer Goods	0.017	3.045	0.000	0.010	32.638
Resources	0.005	0.312	0.006	1.665*	-8.917
Utilities	0.018	3.568***	-0.026	-2.723***	39.703
Industry	0.008	0.759	0.000	0.016	-1.372
Service	0.009	0.780	-0.004	-0.344	-1.300

Even in the presence of the market factor, most of the previous discussed factor loadings for *SMB*, *HML*, and *WML* remain stable as regards their sign with only minor variations in magnitude. In case of Austria, we even find that the factor loadings for *SMB* and *WML* become significant when we run bivariate regressions which include the market factor as well. This implies that, in case of Austria, both *SMB* and *WML* contain incremental information over the market factor, but not necessarily stand alone.

As a whole, the additional information contained on average in the risk factors *HML*, *SMB*, and *WML* is also reflected in a considerable increase in the adjusted R^2 figures, implying that in most cases neither one of the factors nor *MRF* should be omitted. Especially the incremental information of *SMB* should not be denied. This holds in particular from a European stock market integration perspective, given that the slope coefficient for *SMB* in the Eurozone is statistically significant at the 1% significance level.

Our findings depicted in Table 9 also highlight the degree of sensitivity of each industry with respect to business cycle movements. While some industries are very vulnerable to economic fluctuations, other sectors are relatively immune to economic swings. For instance, the relative inelasticity of *non-cyclical consumer goods* reflected by the zero factor loadings for *SMB* and *WML* may serve as an indicator for a certain degree of immunity to variations in economic activities. On the other hand, the findings for *utilities*, *financials*, *information technology*, and *general industries* show a potentially higher sensitivity and vulnerability to economic swings. Overall, however, the factor loadings are relatively small in magnitude, indicating that the return to the factors adds only a small, but still considerable and thus not deniable, portion to the overall future GDP growth rate.

5.3 Fama and French (1993) Factor Regressions

In a subsequent step, we run a regression for the Fama and French (1993) 3FM which contains simultaneously *HML* and *SMB* next to the market factor. The results are depicted in Table 10 for the country regressions and Table 11 for the industry analyses. Focusing on the joint

slope coefficients of the risk factors, this horse-race provides insights on the incremental power contained in one factor relative to the market risk premium and the other factor.

Table 10: Multiple Regressions of GDP Growth Conditional on Past Fama and French (1993) Factor Returns per Country - 4 Quarter Lag

$$GDP_{growth(t,t+4)} = \alpha + \beta MRF_{t-4,t} + hHML_{t-4,t} + sSMB_{t-4,t} + \epsilon_{t,t+4}$$

In the regression notation, MRF is the market risk premium in each country/the Eurozone. The risk free rate is given by the one-month ecu deposit quoted in London. The regressions use annually rebalanced HML, and SMB portfolios. HML is the annual return on a portfolio that is long on high book-to-market stocks and short on low book-to-market securities, holding size and momentum characteristics of the portfolio constant. SMB is the annual return on a portfolio that is long on small capitalization stocks and short on big capitalization securities, holding book-to-market and momentum characteristics of the portfolio constant. GDP is calculated as the continuously compounded growth rate in each country/the Eurozone. The GDP is seasonally adjusted. *T*-statistics are corrected for heteroscedasticity and autocorrelation, up to three lags, using the Newey and West (1987) estimator. The adjusted R^2 is corrected for degrees of freedom. *, **, and *** are used as indicators of statistical significance at, respectively, the 10 %, 5%, and 1% significance level.

Country	MRF		HML		SMB		Adj. R^2 (%)
	Slope	T-Statistics	Slope	T-Statistics	Slope	T-Statistics	
Austria	0.042	6.322***	0.024	3.844***	-0.006	-0.873	78.668
Belgium	0.009	0.728	0.000	0.003	0.024	1.147	3.365
Finland	0.040	7.122***	-0.011	-0.519	-0.004	-0.188	45.476
France	0.009	0.824	0.003	0.202	0.022	1.261	3.982
Germany	0.004	0.253	-0.041	-2.324**	0.005	0.448	9.278
Greece	-0.012	-2.781***	0.001	0.129	0.029	2.814***	22.368
Ireland	0.019	0.447	-0.038	-2.054**	0.035	2.190***	5.040
Italy	-0.012	-0.807	0.024	0.795	0.037	1.573	3.156
Netherlands	0.033	3.068***	-0.020	-1.742*	0.014	1.360	22.776
Portugal	-0.007	-0.373	-0.005	-0.246	0.015	0.914	-4.975
Spain	0.004	0.411	-0.004	-0.474	0.001	0.176	-4.412
Denmark	0.005	0.359	-0.030	-2.074**	0.017	1.386	9.692
Sweden	0.005	0.604	0.015	2.168**	-0.006	-0.522	16.240
United Kingdom	-0.010	-0.654	0.029	1.659*	-0.013	-1.205	1.380
Norway	-0.001	-0.037	0.025	0.639	0.046	1.294	-1.953
Switzerland	-0.008	-0.634	0.005	0.485	0.003	0.369	-4.561
Eurozone	-0.003	-0.219	0.008	0.304	0.065	3.093***	23.394

For most countries and industries the sign remains relatively stable for both *HML* and *SMB*, even in the presence of the market and the other risk factor. Yet, interestingly, country findings suggest that the information contained in *SMB* appears to become somewhat redundant given that we are left with smaller absolute loading factors, of which four out of seven, relative to the bivariate regressions, have lost their statistical significance. On the other hand, the incremental explanatory power of *HML* remains fairly stable as reflected in the seven (out of 17) statistically significant factor loadings. However, when looking at the industry regressions, we face a flipped picture. The magnitude and statistical significance for the *SMB* loadings remains nearly stable, while the number of statistical significant slope coefficients for *HML* drops from three to only one when compared to the bivariate regressions.

Table 11: Multiple Regressions of GDP Growth Conditional on Past Fama and French (1993) Factor Returns per Industry - 4 Quarter Lag

$$GDP_{growth(t,t+4)} = \alpha + \beta MRF_{t-4,t} + hHML_{t-4,t} + sSMB_{t-4,t} + \epsilon_{t,t+4}$$

In the regression notation, MRF is the market risk premium in each industry. The risk free rate is given by the one-month eu deposit quoted in London. The regressions use annually rebalanced HML, and SMB portfolios. HML is the annual return on a portfolio that is long on high book-to-market stocks and short on low book-to-market securities, holding size and momentum characteristics of the portfolio constant. SMB is the annual return on a portfolio that is long on small capitalization stocks and short on big capitalization securities, holding book-to-market and momentum characteristics of the portfolio constant. GDP is calculated as the continuously compounded growth rate in each industry. The GDP is seasonally adjusted. T -statistics are corrected for heteroscedasticity and autocorrelation, up to three lags, using the Newey and West (1987) estimator. The adjusted R^2 is corrected for degrees of freedom. *, **, and *** are used as indicators of statistical significance at, respectively, the 10 %, 5%, and 1% significance level.

Country	MRF		HML		SMB		Adj. R^2 (%)
	Slope	T-Statistics	Slope	T-Statistics	Slope	T-Statistics	
Basic Industries	0.012	1.579	0.005	0.483	-0.002	-0.141	2.211
Cyclical Consumer Goods	0.006	0.505	-0.016	-1.246	0.032	1.666*	7.022
Cyclical Services	0.019	2.181**	0.012	0.558	-0.009	-0.728	6.185
Financials	0.006	0.585	-0.001	-0.090	0.037	2.184**	5.401
General Industries	-0.003	-0.269	0.052	2.674***	0.032	2.471**	24.992
Information Technology	0.023	2.799***	-0.004	-0.969	-0.002	-0.237	40.885
Non-Cyclical Consumer Goods	0.012	1.700*	-0.011	-1.196	0.000	-0.039	35.493
Resources	-0.006	-2.281**	0.000	-0.079	-0.009	-22.929***	88.438
Utilities	0.018	3.183***	-0.020	-1.440	-0.006	-0.459	29.691
Industry	-0.001	-0.064	0.020	0.933	0.040	1.788*	9.590
Service	0.007	0.627	0.017	0.697	0.022	1.064	2.547

Nonetheless, the on average increased adjusted R^2 figures reveal that an inclusion of both *SMB* and *HML* next to the market factor allows for a considerably enhanced explanatory power of the model. This is in line with Liew and Vassalou (2000). Especially, the fairly high coefficients of determination for some industries, such as *general industries*, *information technology*, *non-cyclical consumer goods*, and *utilities*, as well as a relatively high adjusted R^2 for the Eurozone indicate that *SMB*, and *HML* may proxy for state variables that predict macroeconomic growth in the Eurozone. This in turn supports further that equity markets in the Eurozone are to a certain extent integrated.

5.4 Carhart (1997) Factor Regressions

In a final step we run regressions for the proposed Carhart (1997) 4FM, which contains all annually rebalanced risk factors simultaneously. Table 12 depicts the results per country while Table 13 shows the findings per industry. Comparing the results with those found for the Fama and French (1993) regressions, we can see that for most countries and industries the sign and magnitude of the coefficients remain relatively stable. In addition, the inclusion of *WML* adds significant incremental information in the cases of Belgium, Switzerland, and the Eurozone, as well as for *basic industries* and *utilities*. This is underpinned by statistical factor loadings along with increased adjusted R^2 figures. Yet, given again that there is no clear pattern for the sign of the *WML* slope coefficients, it appears that *WML* is either country or industry specific or that *WML* has limited, if at all any, ability to explain future economic growth.

5.5 Sensitivity Analyses

In order to test to what extent our findings are sensitive to the time lag between future GDP growth and past factor returns, we replicated the study using a two year, i.e., eight quarter, rather than a one year, i.e., four quarter, time lag.²⁶

Our findings for *SMB* and *WML* are to a large extent analogous to those that we find for a four quarter lag. Yet, for *HML* we find that the factor returns are predominantly, i.e., overall in 18 out of 28 cases, negatively related to future growth in GDP. In general, as opposed to the four lagged regressions, the findings are less robust and the factors appear to contain slightly less information on future macroeconomic activities. This may indicate that either factor returns are not very persistent or, more likely, that equity markets cannot anticipate future events over

²⁶Due to space considerations the results are not presented here. They can, however, be requested from the authors.

Table 12: Multiple Regressions of GDP Growth Conditional on Past Carhart (1997) Factor Returns per Country - 4 Quarter Lag

$$GDP_{growth(t,t+4)} = \alpha + \beta MRF_{t-4,t} + hHML_{t-4,t} + sSMB_{t-4,t} + wWML_{t-4,t}\epsilon_{t,t+4}$$

Country	MRF		HML		SMB		WML		Adj. R^2 (%)
	Slope	T-Statistics	Slope	T-Statistics	Slope	T-Statistics	Slope	T-Statistics	
Austria	0.040	6.133***	0.028	3.612***	-0.003	-0.594	0.019	1.180	79.159
Belgium	0.000	0.013	0.014	0.527	0.022	1.281	0.036	2.376**	10.328
Finland	0.040	8.186***	-0.011	-0.495	-0.004	-0.188	0.002	0.055	43.726
France	0.009	0.882	0.002	0.154	0.021	1.251	-0.003	-0.175	2.573
Germany	0.007	0.601	-0.038	-2.296**	0.003	0.308	-0.050	-1.554	14.052
Greece	-0.011	-2.516**	0.001	0.113	0.027	2.211**	-0.002	-0.557	16.683
Ireland	0.030	0.698	-0.041	-1.838*	0.025	1.484	-0.022	-1.142	4.453
Italy	-0.009	-0.649	0.026	0.906	0.043	1.741*	-0.036	-1.213	4.036
Netherlands	0.034	3.516***	-0.027	-2.413**	0.016	1.544	-0.020	-0.949	23.837
Portugal	-0.014	-0.736	-0.029	-0.875	0.021	0.956	-0.058	-1.448	3.964
Spain	0.005	0.455	-0.005	-0.631	0.003	0.310	0.006	0.790	-5.722
Denmark	0.006	0.383	-0.030	-2.090**	0.018	1.254	0.004	0.310	6.713
Sweden	0.005	0.623	0.022	1.631	-0.006	-0.452	0.014	0.663	15.432
United Kingdom	-0.011	-0.696	0.030	1.695*	-0.008	-0.656	0.013	0.996	0.514
Norway	0.011	0.407	0.004	0.100	0.035	0.982	-0.041	-1.363	-1.881
Switzerland	-0.010	-0.862	0.009	0.873	0.020	1.642	0.034	2.192**	4.790
Eurozone	-0.006	-0.452	0.015	0.610	0.076	3.826***	0.042	1.941*	23.60

In the regression notation, MRF is the market risk premium in each country/the Eurozone. The risk free rate is given by the one-month ecu deposit quoted in London. The regressions use annually rebalanced HML, SMB, and WML portfolios. HML is the annual return on a portfolio that is long on high book-to-market stocks and short on low book-to-market securities, holding size and momentum characteristics of the portfolio constant. SMB is the annual return on a portfolio that is long on small capitalization stocks and short on big capitalization securities, holding book-to-market and momentum characteristics of the portfolio constant. WML is the annual return on a portfolio that is long on the best performing stocks of the past year ('winners') and short on the worst performing securities of the previous year ('losers') holding book-to-market and size characteristics of the portfolio constant. GDP is calculated as the continuously compounded growth rate in each country/the Eurozone. The GDP is seasonally adjusted. T-statistics are corrected for heteroscedasticity and autocorrelation, up to three lags, using the Newey and West (1987) estimator. The adjusted R^2 is corrected for degrees of freedom. *, **, and *** are used as indicators of statistical significance at, respectively, the 10 %, 5%, and 1% significance level.

Table 13: Multiple Regressions of GDP Growth Conditional on Past Carhart (1997) Factor Returns per Industry - 4 Quarter Lag

$$GDP_{growth(t,t+4)} = \alpha + \beta MRF_{t-4,t} + hHML_{t-4,t} + sSMB_{t-4,t} + wWML_{t-4,t} \epsilon_{t,t+4}$$

In the regression notation, MRF is the market risk premium in industry. The risk free rate is given by the one-month eu deposit quoted in London. The regressions use annually rebalanced HML, SMB, and WML portfolios. HML is the annual return on a portfolio that is long on high book-to-market stocks and short on low book-to-market securities, holding size and momentum characteristics of the portfolio constant. SMB is the annual return on a portfolio that is long on small capitalization stocks and short on big capitalization securities, holding book-to-market and momentum characteristics of the portfolio constant. WML is the annual return on a portfolio that is long on the best performing stocks of the past year ('winners') and short on the worst performing securities of the previous year ('losers') holding book-to-market and size characteristics of the portfolio constant. GDP is calculated as the continuously compounded growth rate in each industry. The GDP is seasonally adjusted. *T*-statistics are corrected for heteroscedasticity and autocorrelation, up to three lags, using the Newey and West (1987) estimator. The adjusted R^2 is corrected for degrees of freedom. *, **, and *** are used as indicators of statistical significance at, respectively, the 10 %, 5%, and 1% significance level.

Country	MRF		HML		SMB		WML		Adj. R^2 (%)
	Slope	T-Statistics	Slope	T-Statistics	Slope	T-Statistics	Slope	T-Statistics	
Basic Industries	0.016	2.008**	0.009	0.930	0.001	0.062	0.023	2.006**	6.351
Cyclical Consumer Goods	0.004	0.326	-0.012	-0.969	0.034	1.763*	0.016	0.715	6.301
Cyclical Services	0.017	1.890*	0.016	0.674	-0.005	-0.410	0.021	1.459	7.886
Financials	0.007	0.621	-0.004	-0.234	0.036	2.091**	-0.007	-0.519	4.177
General Industries	-0.005	-0.422	0.051	2.534**	0.044	2.095**	0.017	1.021	26.036
Information Technology	0.024	3.287***	-0.004	-1.061	-0.004	-0.418	0.001	0.189	37.982
Non-Cyclical Consumer Goods	0.012	1.682*	-0.012	-1.280	0.000	-0.129	-0.001	-0.245	32.315
Resources	-0.003	-0.878	0.003	1.229	-0.008	-9.049***	0.004	1.286	85.084
Utilities	0.017	3.297***	-0.020	-1.183	-0.003	-0.230	-0.028	-2.694***	41.163
Industry	-0.004	-0.279	0.025	1.086	0.054	2.226**	0.034	1.187	12.274
Service	0.006	0.535	0.023	0.799	0.025	1.199	0.015	0.838	1.849

a very long time horizon. Put differently, given our findings, it appears more likely that equity markets are able to anticipate real economic activity one rather than two years ahead. Under equity market integration considerations, however, the presence of same sign tendencies of the factor loadings for different time lags may support our null hypothesis of integrated market, especially when considering our findings for pan-Eurozone industry regressions.

6 Conclusion

The primary aim of this paper was to test the extent to which the profitability of the trading strategies *HML*, *SMB*, and *WML* can, next to the market factor, be related to future economic growth in the Eurozone. Apart from Liew and Vassalou (2000) little to none research has been done that provides evidence of a relation between *HML*, *SMB*, and *WML* and intuitive economic growth. In addition, there is no study that explicitly focuses on different industries - as opposed to different countries - and investigates the euro area. If there exists a relation between these factor returns and future growth in GDP, then these factors may be considered state variables within the context of Merton's ICAPM (Merton, 1973). Albeit this is also of particular interest under equity market integration considerations, this has not yet been addressed by past studies. We suggest that a long-run integration among financial markets in the Eurozone should primarily be attributed to the economic convergence of the member state of the EMU. In turn, for integration to hold, long-run equilibrium relations among equity returns in the Eurozone and their variances should be explained by innovations of common, i.e., pan-Eurozone, state variables.

Using data for sixteen European countries and eleven different industries aggregated across the Eurozone, we show, in line with Aylward and Glen (2000); Barro (1990); Binswanger (2000a,b, 2004); Fama (1981, 1990); Fischer and Merton (1984); Geske and Roll (1983); Liew and Vassalou (2000); Mullins and Wadhwani (1989); Schwert (1990); Wahlroos and Berglund (1986), and Wasserfallen (1989, 1990) that the market factor may serve as leading indicator for future real economic activities in various countries and industries.

In addition, under the consideration of no transaction cost, we suggest that *SMB* contains strong and robust information with respect to future growth in GDP.²⁷ As expected, and in

²⁷The frequent turnover and rebalancing of the portfolios causes transaction costs that we do not consider in this study. These transaction costs in turn consume some of the returns gained. This holds especially for the not very persistent momentum strategy, which results in higher profits if the portfolio is turned over more frequently. In general, *HML* and *SMB* strategies are cheaper to implement than *WML* strategies, because they generate lower transaction costs based on their persistency.

line with Liew and Vassalou (2000) and Perez-Quiros and Timmermann (2000), the relation is positive, indicating that small capitalization firms are better able to prosper than big capitalization stocks whenever strong economic growth is expected. This holds especially for the country regressions. As regards industries, the explanatory power of *SMB* appears to some extent to be industry specific. This is reflected in varying signs of the slope coefficients. Yet, we report strong and robust incremental information content of *SMB* with respect to future real economic activity for the *cyclical consumer goods*, the *financials*, and *general industries* sector. For all other industries the relation is weaker. Overall, the predictive ability of *SMB* is to a large extent independent of any information contained in the market factor.

Noteworthy is the fact that the incremental information content of *SMB* about future GDP growth seems to vanish in various countries, once we include *HML* into our analyses. This, yet, does not hold for the industries considered. In general, *HML* appears to be mainly positively related to future macroeconomic growth. This is again in accordance with Liew and Vassalou (2000). Our findings imply that high book-to-market firms are more likely to thrive as we expect an economic upturn. Besides, we document some support to Lettau and Ludvigson (2001), who suggest that *HML* is particularly sensitive to bad news in bad times.

For *WML* we fail to find a clear and robust pattern for the relation between the factor returns and future growth in GDP. It appears that *WML* is either country or industry specific or that the return to this portfolio strategy has limited, if at all any, ability to explain future macroeconomic growth. One explanation for the low level of information content might be that investors tend to mistakenly project a continuation of abnormal profit levels long periods into the future. This, however, is not in accordance with real economic activity and firms' fundamentals. Successful firms become overvalued and unsuccessful ones become undervalued and the market reacts inefficiently. The market develops a false belief that a few positive or negative events cause a run that will persist for long periods into the future. The market is wrong as past success is not able to project prolonged future success. Consequently, no clear traceable pattern between *WML* and future real economic activity may be detected.

From a theoretical perspective, the findings of this study indicate that *SMB*, and to a lesser extent *HML*, may serve as state variables that predict future changes in the investment opportunity set in context of the Intertemporal Capital Asset Pricing Model (Merton, 1973). In addition, from an equity market integration perspective, the findings for the industry regressions also reveal that equity markets across the Eurozone are to a certain extent integrated as some of the factor returns constructed across pan-Eurozone industries allow for the com-

mon prediction of economic growth in the euro area. The accompanying statistics also suggest that these relations between factor returns and future growth in GDP are not simply due to pure chance. In addition, given that industry factors contain incremental information as regards future real economic activities may also suggest that industry characteristics have gained power relative to country effects in pricing equity. This underpins the findings of Baca et al. (2000); Brooks and Catao (2000); Cavaglia and Moroz (2002); Cavaglia et al. (2000); Campa and Fernandes (2006); Diermeier and Solnik (2001); Ferreira and Gama (2005); Flavin (2004); Isakov and Sonney (2004); L'Her et al. (2002); Moerman (2004, 2008); Taing and Worthington (2005); Urias et al. (1998), and Wang et al. (2003).

An integration of stock markets across the Eurozone implies that equity investors with physical presence in the EMU need to decide on whether to invest outside the EMU to enhance their mean-variance frontier in line with modern portfolio theory (Markowitz, 1952), or, alternatively, to find means to diversify EMU wide if they are reluctant to invest outside the Eurozone. The latter may, for instance, be achieved by diversifying across selected pan-Eurozone industries rather than countries.

Further research may address the concern of using linear regression models to test the relation between the different trading strategies and future GDP growth. For instance, using panel data for 25 countries, Henry et al. (2004) argue that there is strong evidence to suggest that a linear regression model would be inaccurate and would probably provide misleading inference by relating stock market returns to economic output. They remark that different states of the economy produce asymmetric output patterns, i.e., marginal output growth recovers more strongly after a recession than marginal output declines after a boom. In particular, they denote that stock returns are most useful in predicting economic output when an economy is in a recession. Though, Henry et al. (2004) do not employ the same explanatory variables as used in this study, running a switching regression approach that accounts for different states of the economy, may provide further insights into the information content of *HML*, *SMB*, and *WML* with respect to future growth in GDP.

In order to find out more about the information content of *HML*, *SMB*, and *WML* regarding future growth in the economy, one could run further regressions with additional explanatory variables. For instance, Liew and Vassalou (2000) suggest to use the return on a market portfolio (*MRF*), a dividend yield, short-term interest rates, term spreads (e.g., ten year government yield minus the yield on a treasury bill or the call money rate), and the industrial production as indicators for the business cycle. Liew and Vassalou (2000) show that there exists some

overlap in the information content of *HML*, *SMB* and the business cycle variables. This is also corroborated by Petkova (2006). It would perhaps be interesting to see whether this holds for industry corrected models as well. Yet, it should not necessarily be the aim to propose a new model for the prediction of future growth in the macroeconomy. The question is more, whether *SMB*, *HML*, or any of the other trading strategies or state variables may contain some additional information above the market factor itself.

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A Appendix

Table I: VIF per Country and Total - Annually Rebalanced

This table reports the variance inflation factor (VIF) for all risk factors per country and the total European market, i.e., the Eurozone, European Union, and Europe as a whole. The VIF is defined as

$$VIF(\hat{\beta}_i) = \frac{1}{1 - R_i^2}$$

and is estimated by regressing each of the variables on the remaining three using all observations available per country. The countries are clustered along three dimensions. The first group comprises those countries that belong to the Eurozone. The second cluster represents countries of the European Union that do not belong to the Eurozone. The last cluster contains European countries that neither belong to the Eurozone nor the European Union. The results are based on annually rebalanced HML, SMB, and WML portfolios using monthly observations. MRF denotes the market risk factor. HML is the return on a portfolio that is long on high book-to-market stocks and short on low book-to-market securities, holding size and momentum characteristics of the portfolio constant. SMB is the return on a portfolio that is long on small capitalization stocks and short on big capitalization securities, holding book-to-market and momentum characteristics of the portfolio constant. WML is the return on a portfolio that is long on the best performing stocks of the past year ('winners') and short on the worst performing securities of the previous year ('losers') holding book-to-market and size characteristics of the portfolio constant.

	MRF	HML	SMB	WML
Austria	1.140	1.253	1.070	1.060
Belgium	1.381	1.216	1.288	1.615
Finland	1.456	4.264	4.877	1.010
France	1.244	1.152	1.192	1.268
Germany	1.368	1.646	1.379	1.128
Greece	1.304	2.322	2.074	1.073
Ireland	1.280	1.322	1.410	1.366
Italy	1.361	1.368	1.048	1.062
Luxembourg*	-	-	-	-
Netherlands	1.120	1.272	1.098	1.255
Portugal	1.276	3.665	4.201	1.783
Spain	1.780	1.057	1.740	1.368
Denmark	1.311	1.091	1.464	1.255
Sweden	1.220	3.456	1.087	3.089
United Kingdom	1.138	1.055	1.236	1.191
Norway	1.646	1.825	1.360	1.395
Switzerland	1.235	1.251	1.950	1.970
Eurozone	1.186	1.041	1.243	1.100

* Not sufficient data available

Table II: VIF per Industry (Eurozone) - Annually Rebalanced

This table reports the variance inflation factor (VIF) for all risk factors per industry across the Eurozone. The VIF is defined as

$$VIF(\hat{\beta}_i) = \frac{1}{1 - R_i^2}$$

and is estimated by regressing each of the variables on the remaining three using all observations available per industry. The results are based on annually rebalanced HML, SMB, and WML portfolios using monthly observations. MRF denotes the market risk factor. HML is the return on a portfolio that is long on high book-to-market stocks and short on low book-to-market securities, holding size and momentum characteristics of the portfolio constant. SMB is the return on a portfolio that is long on small capitalization stocks and short on big capitalization securities, holding book-to-market and momentum characteristics of the portfolio constant. WML is the return on a portfolio that is long on the best performing stocks of the past year ('winners') and short on the worst performing securities of the previous year ('losers') holding book-to-market and size characteristics of the portfolio constant.

	MRF	HML	SMB	WML
Basic Industries	1.140	1.197	1.488	1.210
Cyclical Consumer Goods	1.290	1.175	1.089	1.130
Cyclical Services	1.108	1.135	1.064	1.063
Financials	1.111	1.197	1.130	1.133
General Industries	1.250	1.090	2.558	2.390
Information Technology	1.264	4.073	3.280	1.824
Non-Cyclical Consumer Goods	1.308	1.340	1.602	1.152
Non-Cyclical Services*	-	-	-	-
Resources	1.455	1.294	2.432	3.289
Utilities	1.024	1.727	1.748	1.011
Industry (aggregate)	1.328	1.025	1.775	1.428
Service (aggregate)	1.062	1.326	1.211	1.210

* Not sufficient data available